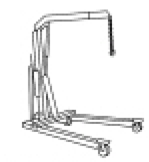
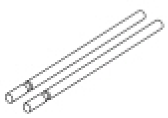
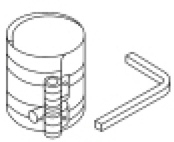
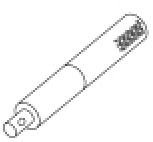
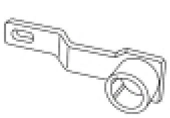
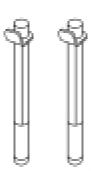
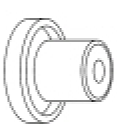
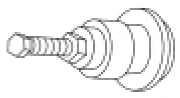
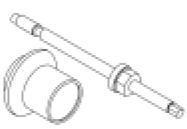
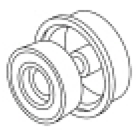
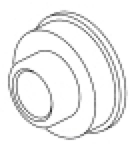
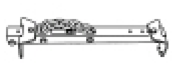
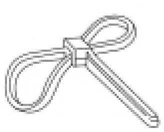
Engine

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST1341-A	2,200# Floor Crane, Fold Away 300-OTC1819E
 ST2806-A	Alignment Pins, Cylinder Head 303-1040 (SR-015486)
 ST1376-A	Compressor, Piston Ring 303-D032 (D81L-6002-C) or equivalent
 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST1335-A	Holding Tool, Crankshaft 303-448 (T93P-6303-A)
 ST1337-A	Installer, Connecting Rod 303-442 (T93P-6136-A)
 ST2212-A	Installer, Differential Bearing Cone 205-142 (T80T-4000-J)

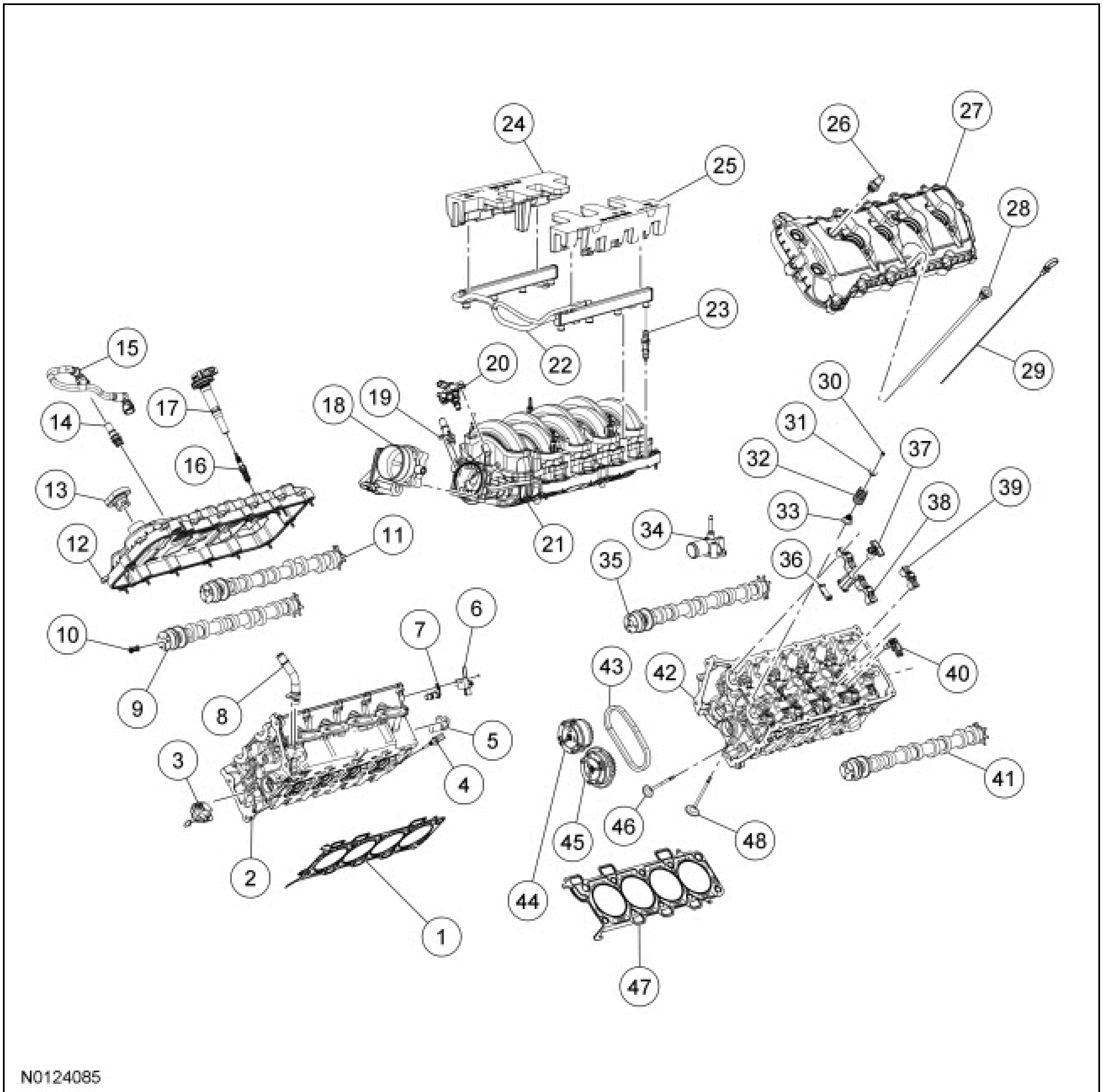
 ST1326-A	Installer, Front Cover Oil Seal 303-335 (T88T-6701-A2 plate only)
 ST3249-A	Installer, Front Crank Seal and Damper 303-1531
 ST2980-A	Installer, Rear Main Seal 303-1250
 ST2983-A	Installer, Spark Plug Tube Seal 303-1247/2
 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent

Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A
Motorcraft® SAE 5W-50 Full Synthetic Motor Oil XO-5W50-QGT	WSS-M2C931-C
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Orange Concentrated Antifreeze/Coolant VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D

Motorcraft® Silicone Brake Caliper Grease and Dielectric Compound XG-3-A	ESE-M1C171-A
Motorcraft® Silicone Gasket and Sealant TA-30	WSE-M4G323-A4

Engine — Upper End

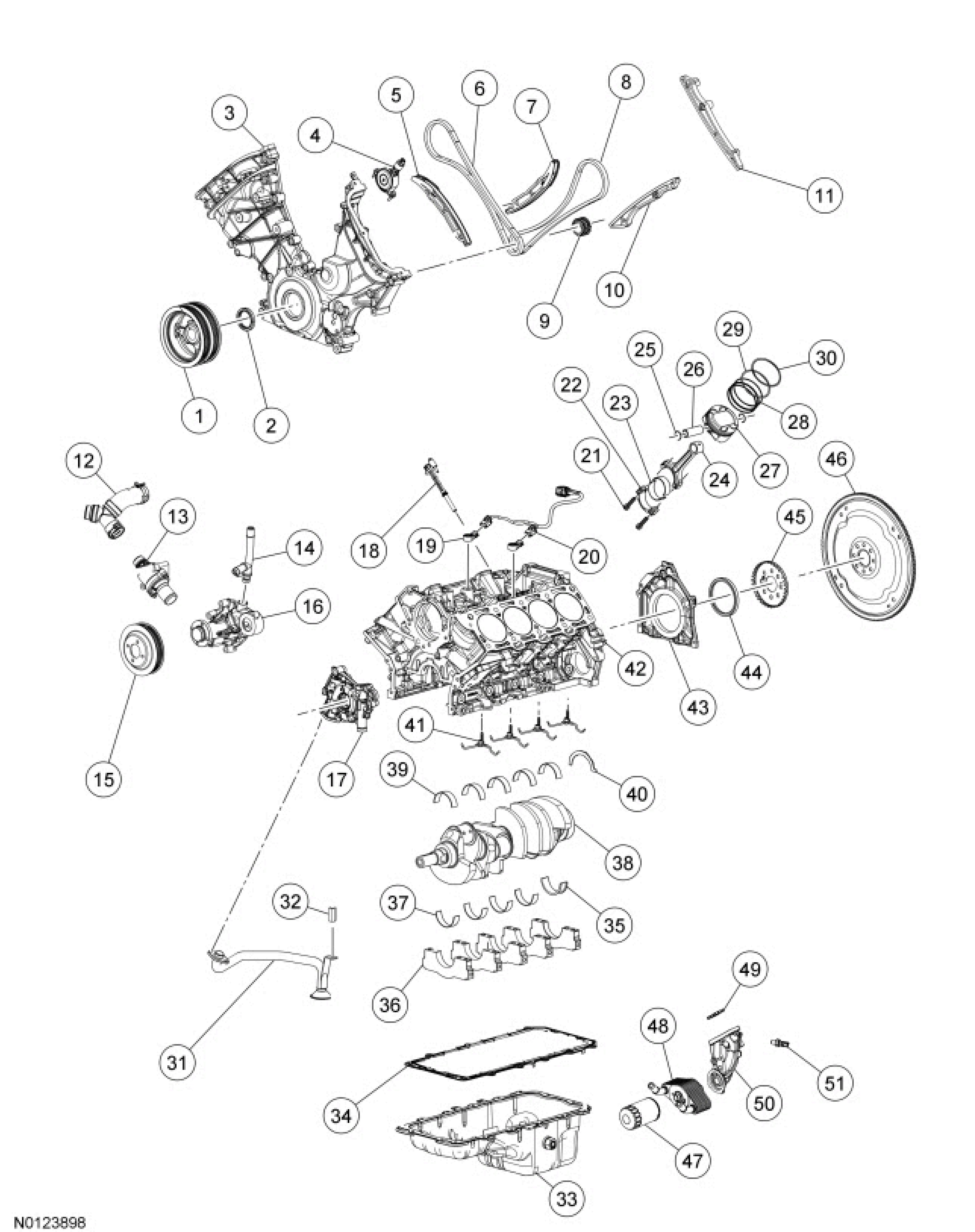


Item	Part Number	Description
1	60516051	RH cylinder head gasket
2	60496049	RH cylinder head
3	6L2666L266	Primary timing chain tensioner (2 required)
4	6G0046G004	Cylinder Head Temperature (CHT) sensor
5	6B2886B288	Intake Camshaft Position (CMP) sensor (2 required)
6	1880118801	Radio capacitor (2 required)
7	12K07312K073	Exhaust CMP sensor (2 required)

8	1869618696	Heater supply tube
9	62506250	RH exhaust camshaft
10	6C6836C683	Camshaft engine oil filter (4 required)
11	62506250	RH intake camshaft
12	65826582	RH valve cover
13	67666766	Oil filler cap
14	6A6666A666	PCV valve
15	65786578	Crankcase vent hose
16	1240512405	Spark plug (8 required)
17	1202912029	Ignition coil on plug (8 required)
18	9F9919F991	Throttle Body (TB)
19	9F6249F624	Heat control inlet tube
20	9C9159C915	Fuel vapor purge valve
21	94249424	Intake manifold
22	9F7929F792	Fuel rail
23	9F5939F593	Fuel injector (8 required)
24	6P0136P013	RH fuel injector insulator shield
25	6P0136P013	LH fuel injector insulator shield
26	67626762	Crankcase vent valve
27	65826582	LH valve cover
28	67546754	Oil indicator tube
29	67506750	Oil indicator
30	65186518	Valve spring retainer key (64 required)
31	65146514	Valve spring retainer (32 required)
32	65136513	Valve spring (32 required)
33	65716571	Valve stem seal (32 required)
34	85948594	Coolant outlet connector
35	62506250	LH intake camshaft
36	6K2976K297	Secondary tensioner shoe (2 required)
37	6K2546K254	Secondary tensioner (2 required)
38	—	Camshaft bearing cap (part of 6049) (2 required)
39	—	Camshaft bearing cap (part of 6049) (16 required)
40	65006500/6564/6564	Hydraulic lash adjuster/camshaft roller follower (32 required)
41	62506250	LH exhaust camshaft
42	60496049	LH cylinder head
43	62686268	Secondary timing chain (2 required)
44	62566256	Intake camshaft Variable Camshaft Timing (VCT) (2 required)
45	62566256	Exhaust camshaft VCT (2 required)
46	65056505	Exhaust valve (16 required)

47	60516051	LH cylinder head gasket
48	65076507	Intake valve (16 required)

Engine — Lower End



Item	Part Number	Description
1	63166316	Crankshaft pulley

2	67006700	Crankshaft front oil seal
3	60196019	Engine front cover
4	6B2976B297	Variable Camshaft Timing (VCT) variable force solenoid (4 required)
5	6K2556K255	RH timing chain tensioner arm
6	62686268	RH primary timing chain
7	6K2556K255	LH timing chain tensioner arm
8	62686268	LH primary timing chain
9	63066306	Crankshaft sprocket
10	6B2746B274	LH timing chain guide
11	6M2566M256	RH timing chain guide
12	85668566	Coolant bypass tee
13	8A5868A586	Thermostat housing
14	1866318663	Heater outlet tube
15	85098509	Coolant pump pulley
16	85018501	Coolant pump
17	66006600	Oil pump
18	6C3156C315	Crankshaft Position (CKP) sensor
19	12A69912A699	KS (2 required)
20	14A41114A411	KS wiring harness
21	62146214	Connecting rod cap bolt (16 required)
22	—	Connecting rod cap (part of 6200) (8 required)
23	62116211	Connecting rod bearing (16 required)
24	62006200	Connecting rod (8 required)
25	61406140	Piston pin retainer (16 required)
26	61356135	Piston pin (8 required)
27	61106110	Piston (8 required)
28	—	Outer oil control ring (part of 6148) (16 required)
29	—	Lower compression ring (part of 6148) (8 required)
30	—	Upper compression ring (part of 6148) (8 required)
31	66226622	Oil pump screen and pickup tube
32	N806180N806180	Oil pump screen and pickup tube spacer
33	66756675	Oil pan
34	67106710	Oil pan gasket
35	—	Crankshaft main bearing cap (part of 6010) (5 required)
36	6D3096D309	Crankshaft lower main bearing
37	6D3096D309	Crankshaft lower main bearing (4 required)
38	63036303	Crankshaft
39	6D3096D309	Crankshaft upper main bearing (5 required)
40	6A3416A341	Thrust washer

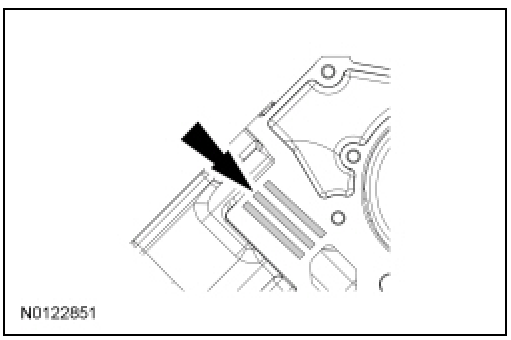
41	6K8686K868	Piston cooling jets (4 required)
42	60106010	Engine block
43	6K3186K318	Crankshaft rear seal retainer plate
44	67016701	Crankshaft rear seal
45	12A22712A227	Crankshaft sensor ring
46	63756375	Flexplate
47	67316731	Oil filter
48	6A6426A642	Oil cooler
49	68406840	Oil filter adapter gasket
50	68816881	Oil filter adapter
51	92789278	Engine Oil Pressure (EOP) switch

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan, can cause engine failure.

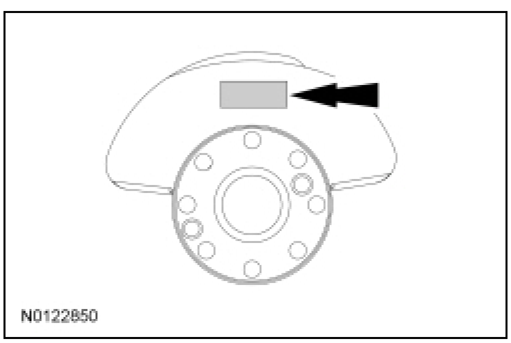
NOTICE: During engine assembly do not allow the piston connecting rods to come in contact with the piston cooling jets or engine damage may occur.

NOTE: *During engine assembly it may become necessary to check bearing clearances and end play. For additional information, refer to Section 303-00.*

1. Record the main bearing code found on the front of the engine block.



2. Record the main bearing code found on the rear of the crankshaft.



3. **NOTE:** *This chart is for selecting main bearings 1 and 5 only, the remaining bearings will be selected using a different chart in the next step.*

   Using the data recorded earlier and the Bearing Select Fit Chart, Standard Bearings, determine the required bearing grade for main bearings 1 and 5.
   - Read the first letter of the engine block main bearing code and the first letter of the crankshaft main bearing code.

- Read down the column below the engine block main bearing code letter and across the row next to the crankshaft main bearing code letter, until the 2 intersect. This is the required bearing grade(s) for the No. 1 crankshaft main bearing.
- As an example, if the engine block code letter is "F" and the crankshaft code letter is "P", the correct bearing grade for this main bearing is a "1" for the upper bearing and a "2" for the lower bearing.
- Repeat the above steps using the fifth letter of the block and crankshaft codes to select the No. 5 bearing.

MAINS 1 & 5

UPPER/LOWER

MINIMUM BLOCK DIA

72.400 .401 .402 .403 .404 .405 .406 .407 .408 .409 .410 .411 .412 .413 .414 .415 .416 .417 .418 .419 .420 .421 .422 .423 .424

MAXIMUM CRANKSHAFT DIA

67.505 67.504 67.503 67.502 67.501 67.500 67.499 67.498 67.497 67.496 67.495 67.494 67.493 67.492 67.491 67.490 67.489 67.488 67.487 67.486 67.485 67.484 67.483 67.482 67.481

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
X	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2
W	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3
V	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3
U	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3
T	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3
S	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3
R	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3
Q	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3
P	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3
O	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3
N	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3
M	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3
L	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3
K	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3
J	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3
I	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
H	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
G	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
F	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
E	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
D	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
C	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
B	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
A	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3

N0124043

4. **NOTE:** *This chart is for selecting main bearings 2, 3 and 4 only.*

   Using the data recorded earlier and the Bearing Select Fit Chart, Standard Bearings, determine the required bearing grade for main bearings 2, 3 and 4.
   - Read the second letter of the engine block main bearing code and the second letter of the crankshaft main bearing code.
   - Read down the column below the engine block main bearing code letter and across the row next to the crankshaft main bearing code letter, until the 2 intersect. This is the required bearing grade(s) for the No. 2 crankshaft main bearing.
   - As an example, if the engine block code letter is "F" and the crankshaft code letter is "P", the correct bearing grade for this main bearing is a "1" for the upper bearing and a "2" for the lower bearing.
   - Repeat the above steps using the third and fourth letters of the block and crankshaft codes to select the No. 3 and No. 4 bearings.

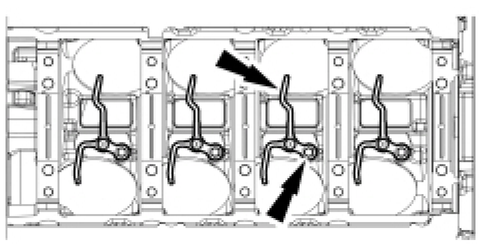
MINIMUM BLOCK DIA

MAINS 2, 3 & 4 UPPER/LOWER		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
MAXIMUM CRANKSHAFT DIA	72.400	.401	.402	.403	.404	.405	.406	.407	.408	.409	.410	.411	.412	.413	.414	.415	.416	.417	.418	.419	.420	.421	.422	.423	.424
X	67.505	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2
W	67.504	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2
V	67.503	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2
U	67.502	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2
T	67.501	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/3	2/2	2/2	2/2
S	67.500	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3
R	67.499	1/1	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3
Q	67.498	1/1	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3
P	67.497	1/1	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3
O	67.496	1/1	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3
N	67.495	1/1	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3
M	67.494	1/1	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3
L	67.493	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3
K	67.492	1/2	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3
J	67.491	1/2	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3
I	67.490	1/2	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3
H	67.489	1/2	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3
G	67.488	1/2	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3
F	67.487	1/2	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3
E	67.486	1/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
D	67.485	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
C	67.484	2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
B	67.483	2/2	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
A	67.482	2/2	2/2	2/2	2/2	2/2	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
	67.481																								

N0124042

NOTE: *Before assembling the cylinder block, all sealing surfaces must be free of chips, dirt, paint and foreign material. Also, make sure the coolant and oil passages are clear.*

5. Install the 4 piston cooling jets and 4 bolts. Tighten in 2 stages.
   - Stage 1: Tighten to 10 Nm (89 lb-in).
   - Stage 2: Tighten an additional 45 degrees.

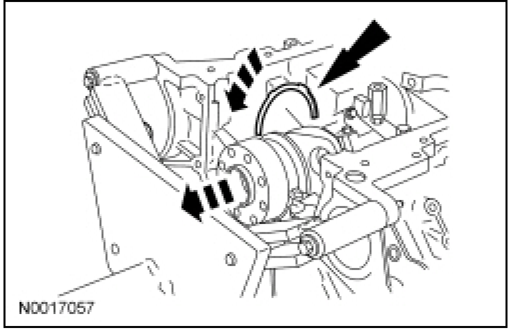


N0122848

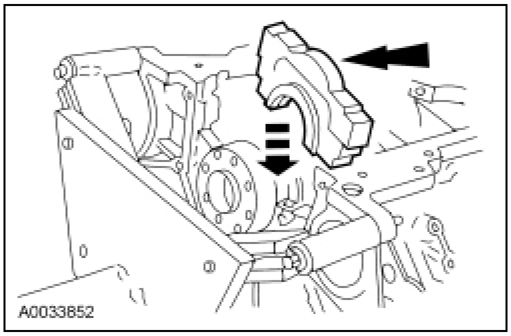
6. Install the crankshaft main bearings.
   - Install the crankshaft upper main bearings into the cylinder block.
   - Install the crankshaft lower main bearings and thrust bearing into the bearing caps.
   - Make sure all oil passages are aligned.
   - Lubricate all main bearings with clean engine oil.
7. **NOTE:** *Lubricate the crankshaft bearing journals with clean engine oil.*

   Install the crankshaft into the cylinder block.
8. **NOTE:** *The oil groove on the thrust washer must face toward the rear of the engine (against the crankshaft thrust surface).*

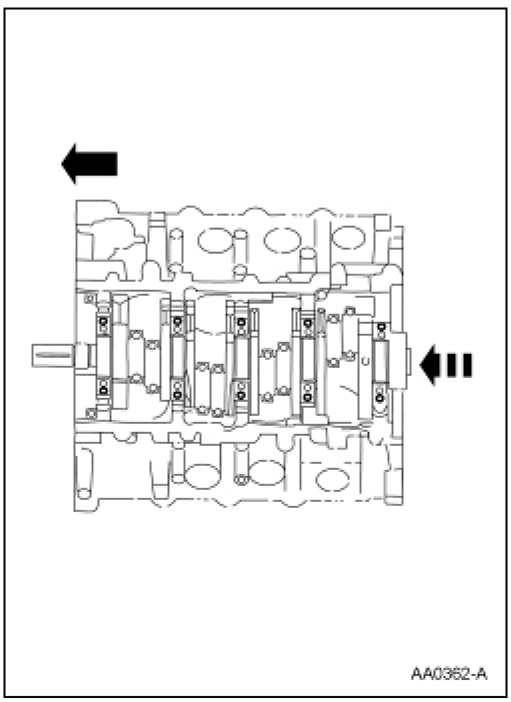
   Push the crankshaft rearward and install the rear crankshaft upper thrust washer at the back of the No. 5 main boss.



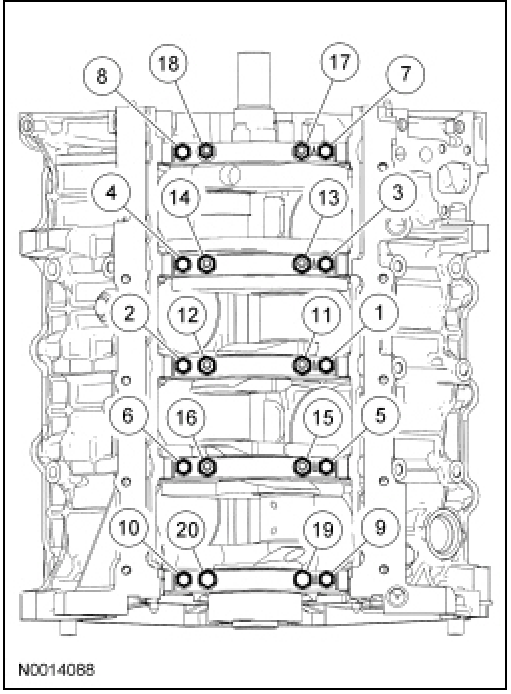
9. Install the main bearing caps and fasteners on the cylinder block and, keeping the caps as square as possible, alternately draw the caps down evenly using the cap fasteners.



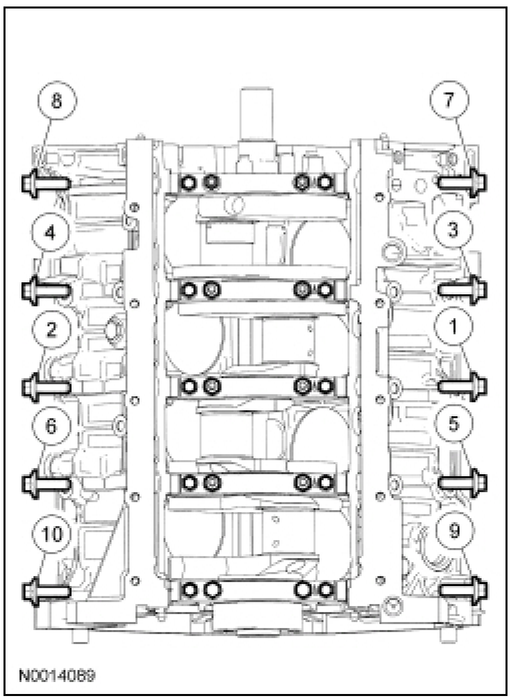
10. Push the crankshaft forward to seat the crankshaft thrust washer. Hold the crankshaft in the forward position.



11. Tighten the vertical main bearing cap fasteners in the sequence shown, in 4 stages.
    - Stage 1: Tighten fasteners 1 through 20 to 20 Nm (177 lb-in).
    - Stage 2: Tighten fasteners 1 through 10 to 40 Nm (30 lb-ft).
    - Stage 3: Tighten fasteners 11 through 20 to 65 Nm (48 lb-ft).
    - Stage 4: Tighten fasteners 1 through 20 an additional 90 degrees.



12. Install the cross-mounted main bearing cap fasteners and tighten in the sequence shown, in 3 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten to 30 Nm (22 lb-ft).
    - Stage 3: Tighten an additional 60 degrees.



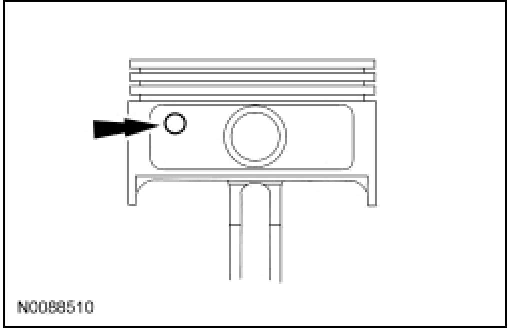
13. Check the crankshaft end play. For additional information, refer to Section 303-00.

14. Check that crankshaft torque-to-turn does not exceed 6 Nm (53 lb-in).

15. Check the piston-to-cylinder block and piston ring clearances. For additional information, refer to Section 303-00.

16. Assemble the pistons and position the piston ring gaps. For additional information, refer to Piston in this section.

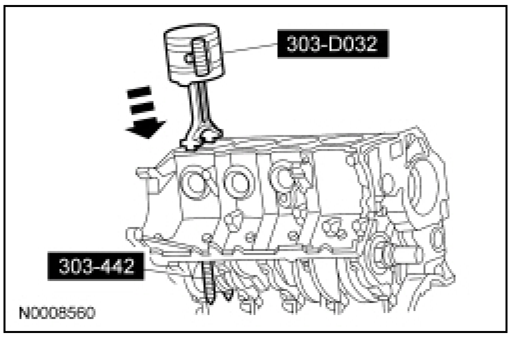
17. Make sure the dimple in the piston faces the front of the engine.



18. **NOTICE: Do not scratch the cylinder walls or crankshaft journals with the connecting rod or engine damage may occur.**

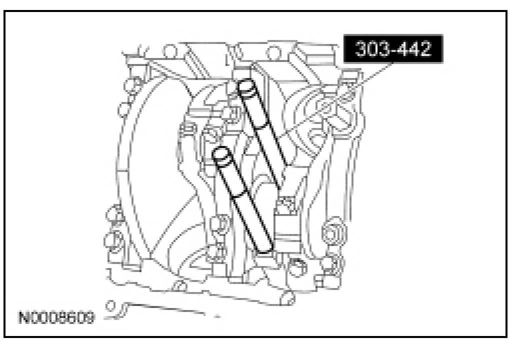
    NOTICE: Do not allow the piston connecting rods to come in contact with the piston cooling jets or engine damage may occur.

    Use the Piston Ring Compressor and the Connecting Rod Installers to install the connecting rod with the upper connecting rod bearing in place.
    - Lubricate the piston and ring with clean engine oil.
    - Lubricate the rod bearings with clean engine oil.



19. **NOTICE: Do not scratch the cylinder walls or crankshaft journals with the connecting rod or engine damage may occur.**

    Once the connecting rod is seated on the crankshaft journal, remove the Connecting Rod Installers.



20. **NOTICE: The rod cap installation must keep the same orientation as marked during disassembly or engine damage may occur.**

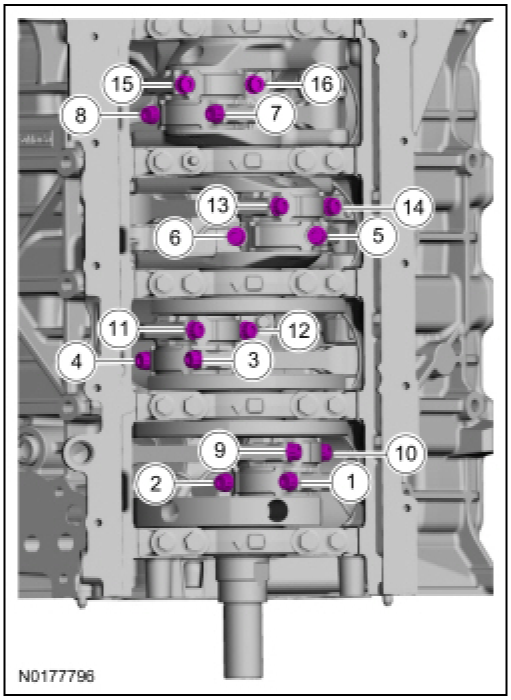
    NOTE: *The connecting rod caps are of the "cracked" design and must mate with the connecting rod ends. Excessive bearing clearance will result if not mated correctly.*

    Position the lower bearing and connecting rod and install the new bolts loosely.

21. Repeat the previous 5 steps for each of the remaining pistons.

22. Tighten the bolts in the sequence shown in 3 stages.
    - Stage 1: Tighten to 20 Nm (177 lb-in).
    - Stage 2: Tighten to 38 Nm (28 lb-ft).

- Stage 3: Tighten an additional 105 degrees.

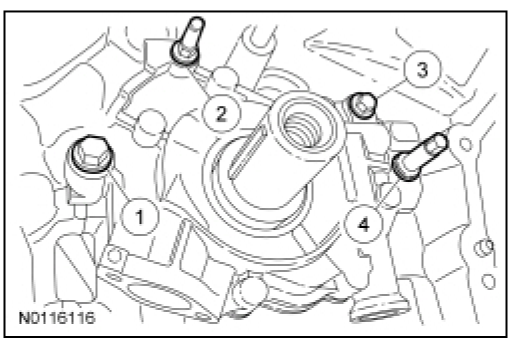


23. Prime the oil pump. Add 2 tablespoons of clean engine oil to the oil pump and rotate the oil pump by hand.

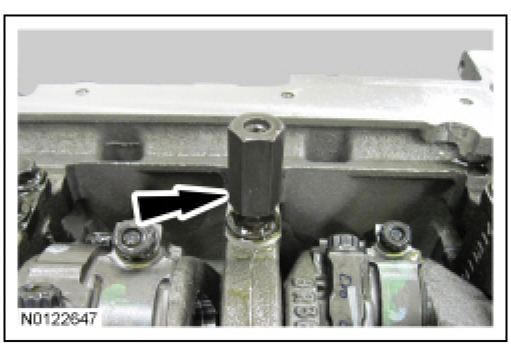
24. Rotate the inner rotor of the oil pump assembly to align the flats on the crankshaft and slip the oil pump over the crankshaft until seated against the block.
    - Rotate the oil pump until the bolt holes are aligned to the block and install the fasteners.

25. **NOTE:** *Oil pump must be held against the cylinder block until all bolts are tightened*.

    Tighten the fasteners in the sequence shown in 3 stages:
    - Stage 1: Hand tighten.
    - Stage 2: Tighten the bolt (1) to 10 Nm (89 lb-in), the stud bolt (2) to 25 Nm (18 lb-ft), the bolt (3) to 10 Nm (89 lb-in) and the stud bolt (4) to 20 Nm (177 lb-in).
    - Stage 3: Tighten the bolt (1) an additional 45 degrees, the stud bolt (2) an additional 75 degrees, the bolt (3) an additional 45 degrees and the stud bolt (4) an additional 60 degrees.

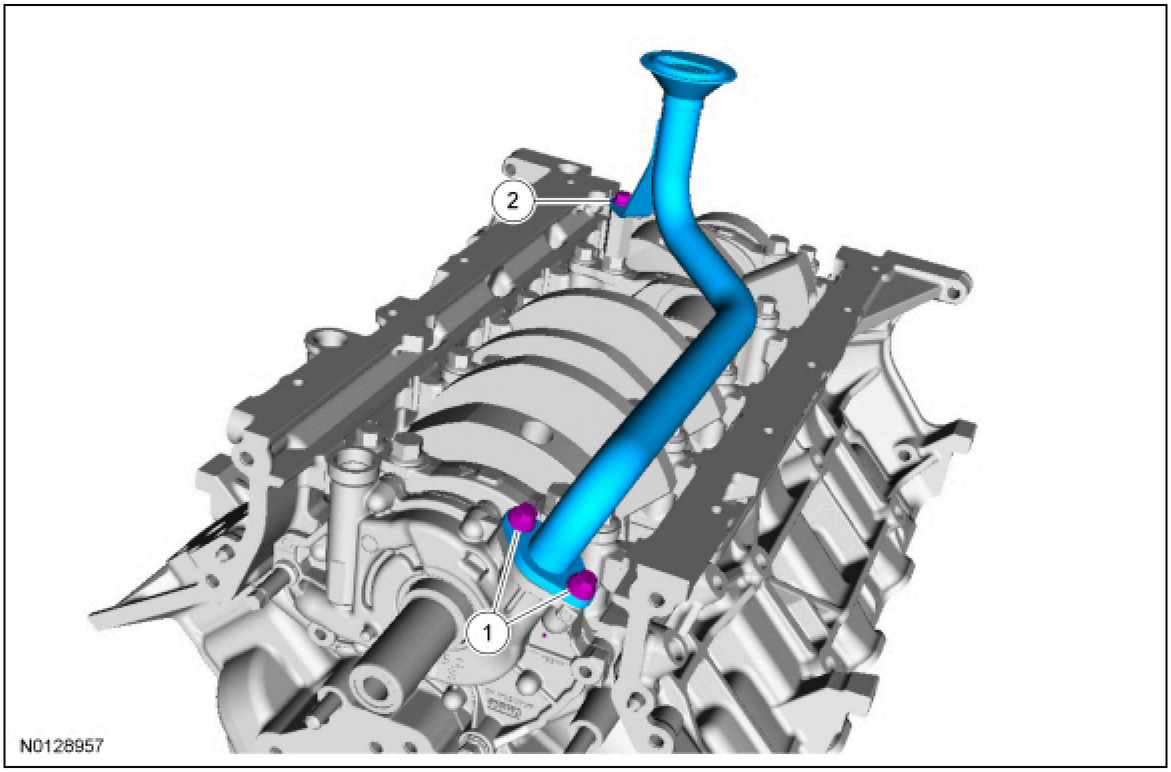


26. Install the oil pump screen and pickup tube spacer.
    - Tighten to 25 Nm (18 lb-ft).



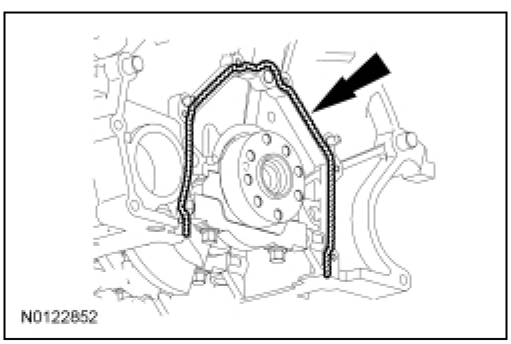
27. **NOTE:** *The 2 oil pump screen and pickup tube-to-oil pump bolts must be tightened prior to tightening the oil pump screen and pickup tube-to-spacer bolt.*

    Using a new O-ring seal, install the oil pump screen and pickup tube and the 3 bolts.
    1. Tighten the 2 oil pump screen and pickup tube-to-oil pump bolts in 2 stages.
        - Stage 1: Tighten to 10 Nm (89 lb-in).
        - Stage 2: Tighten an additional 45 degrees.
    2. Tighten the oil pump screen and pickup tube-to-spacer bolt to 25 Nm (18 lb-ft).

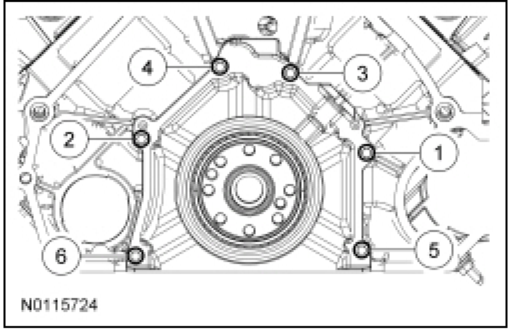


28. **NOTE:** *If the rear crankshaft seal retaining plate is not secured within 4 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure may cause future oil leaks.*

    Apply a 3.75 mm (0.147 in) bead of silicone gasket and sealant to the rear crankshaft seal retainer mating surface on the engine block.



29. Install the crankshaft rear seal retainer plate and the 6 bolts, tighten in the sequence shown in 2 stages.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

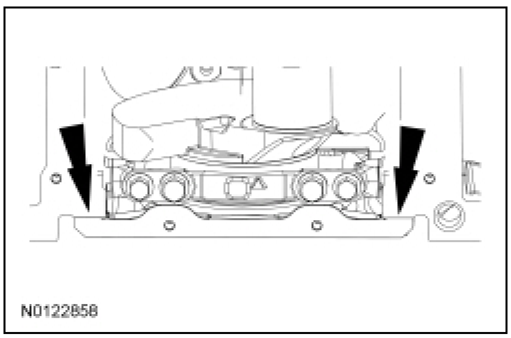


30. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

    Inspect the oil pan and engine sealing surfaces. Clean the mating surfaces of the engine and oil pan with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

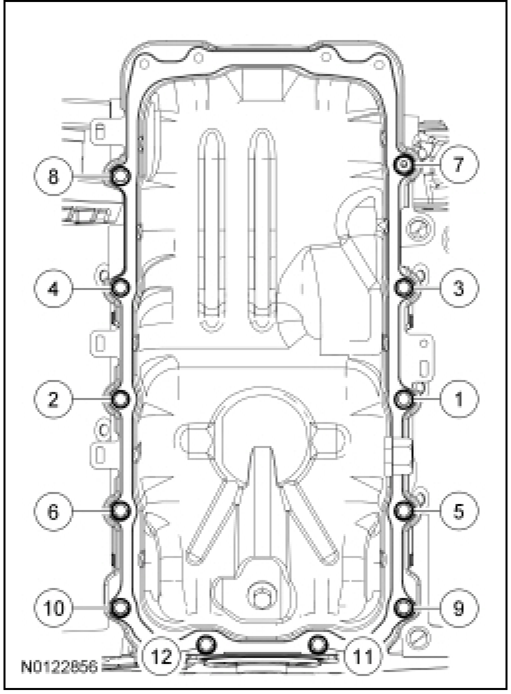
31. **NOTE:** *If the oil pan is not installed and the fasteners tightened within 5 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage. If this timing cannot be met, tighten fasteners 7, 8, 9 and 10 to 8 Nm (71 lb-in) within 5 minutes of applying the sealer and final torque all of the fasteners within 1 hour of applying the sealer.*

    Apply an 8 mm (0.31 in) bead of silicone gasket and sealant to the crankshaft rear seal retainer plate-to-cylinder block sealing surfaces and the engine front cover-to-cylinder block sealing surfaces.

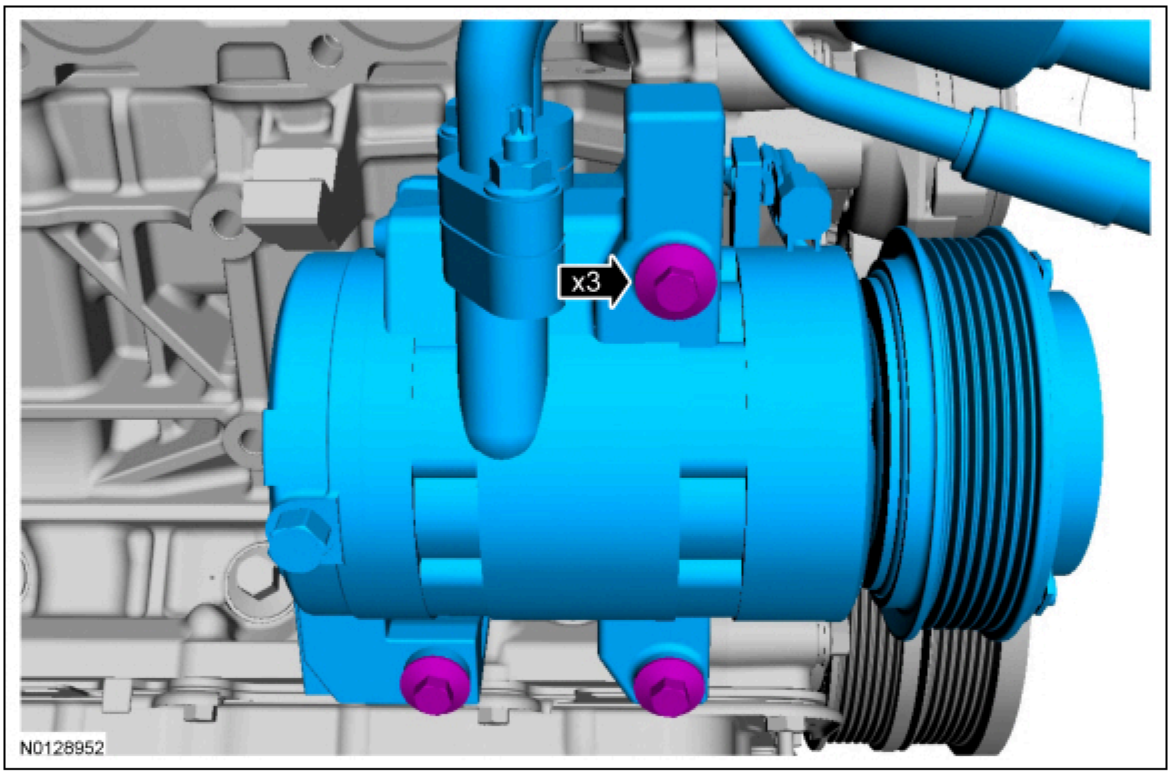


32. **NOTE:** *Fastener location 7 is a stud bolt.*

    Using a new gasket, install the oil pan and tighten the fasteners in sequence in 3 stages.
    - Stage 1: Tighten to 2 Nm (18 lb-in)
    - Stage 2: Tighten to 10 Nm (89 lb-in).
    - Stage 3: Tighten an additional 45 degrees.

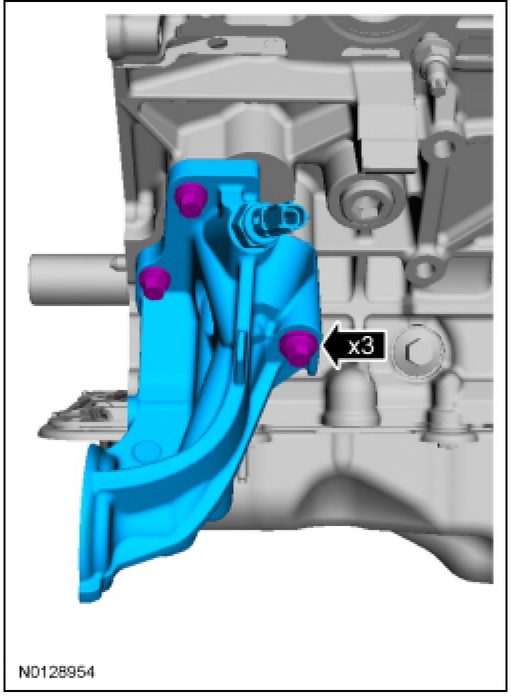


33. Install the A/C compressor and the 3 bolts.
    - Tighten to 25 Nm (18 lb-ft).



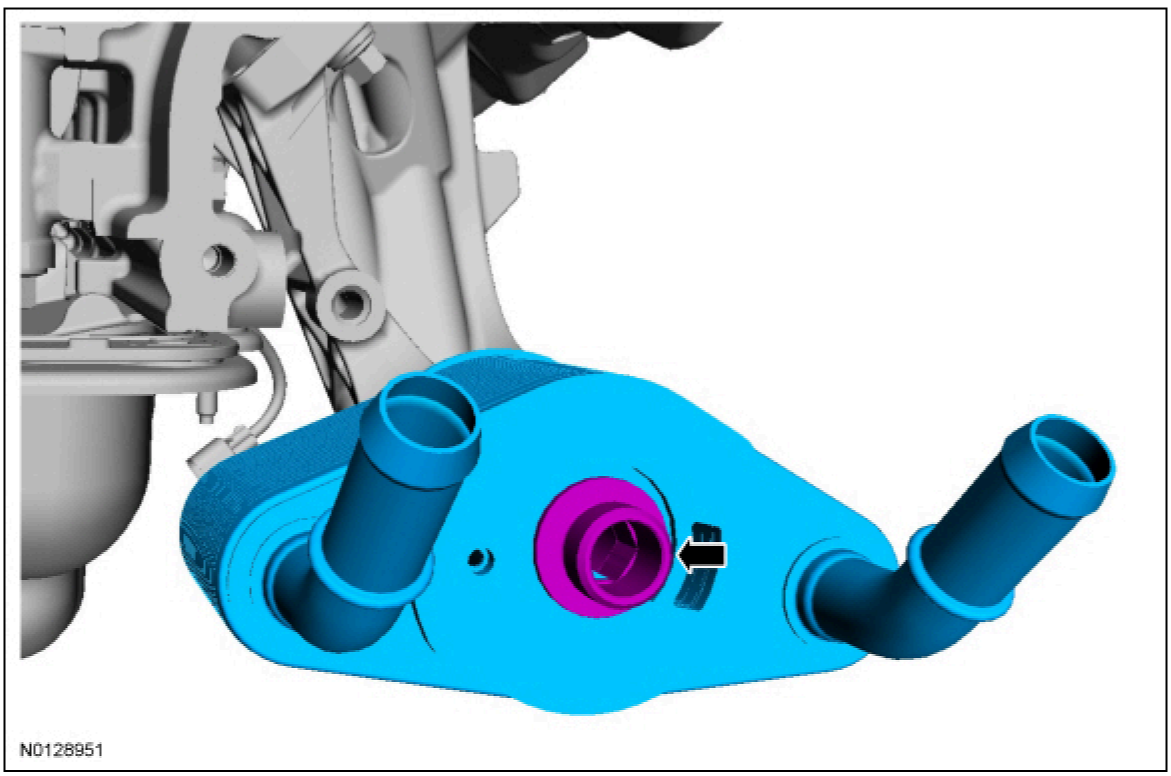
34. **NOTE:** *Install a new oil filter adapter gasket.*

    Using a new gasket, install the oil filter adapter and the 3 bolts. Tighten the bolts in 2 stages.
    - Stage 1: Tighten the bolts to 20 Nm (177 lb-in).
    - Stage 2: Tighten an additional 60 degrees.



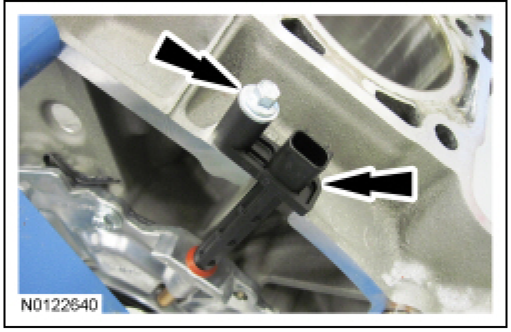
35. **NOTICE: A new oil cooler must be installed or severe damage to the engine can occur.**

    Install a new oil cooler and the threaded fitting.
    - Tighten to 57 Nm (42 lb-ft).

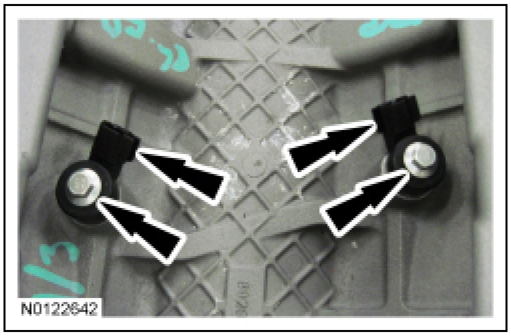


36. **NOTICE: The Crankshaft Position (CKP) sensor must be positioned into the fitting on the crankshaft rear seal retainer plate and be flush against the boss on the engine block before the bolt is installed. If the CKP sensor is installed incorrectly, the CKP sensor can be damaged.**

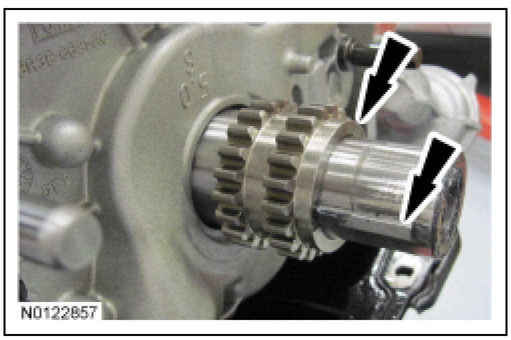
    Position the CKP sensor and install the bolt.
    - Tighten to 10 Nm (89 lb-in).



37. Install the 2 Knock Sensor (KS) and 2 bolts.
    - Tighten to 20 Nm (177 lb-in).



38. Verify the crankshaft keyway is in the 9 o'clock position and install the crankshaft sprocket with the flange facing forward.



39. **NOTICE: Make sure all coolant residue and foreign material are cleaned from the block surface and cylinder bore. Failure to follow these instructions may result in engine damage.**

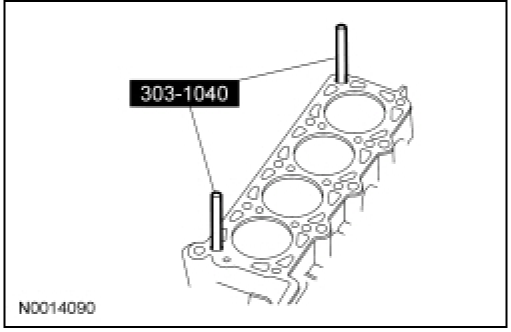
    NOTICE: The use of sealing aids (aviation cement, copper spray and glue) is not permitted. The gasket must be installed dry. Failure to follow these instructions may result in future oil leakage.

    NOTICE: The cylinder head bolts must be discarded and new bolts installed. They are a tighten-to-yield design and cannot be reused.

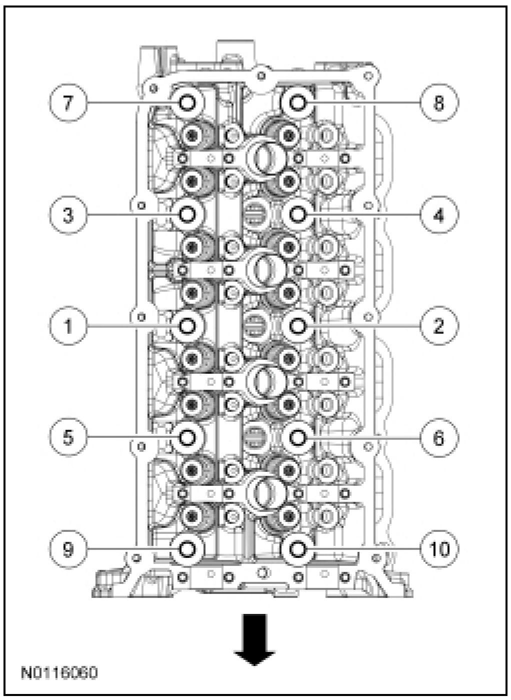
    NOTE: *Do not turn the crankshaft until instructed to do so.*

    NOTE: *LH shown, RH similar.*

    Using the Cylinder Head Alignment Pins, position the cylinder head gaskets and cylinder heads over the dowels and install the cylinder head bolts loosely.



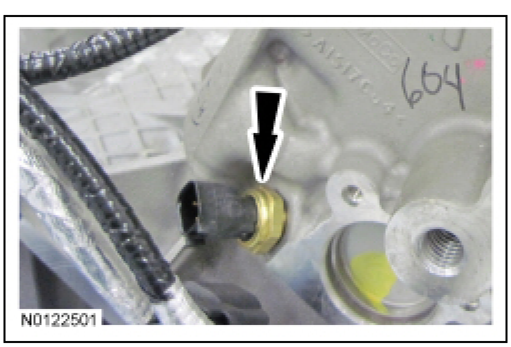
40. Tighten the new RH cylinder head bolts in 7 stages in the sequence shown.
    - Stage 1: Tighten to 25 Nm (18 lb-ft).
    - Stage 2: Tighten to 75 Nm (55 lb-ft).
    - Stage 3: **Loosen** 180 degrees.
    - Stage 4: Tighten to 25 Nm (18 lb-ft).
    - Stage 5: Tighten to 40 Nm (30 lb-ft).
    - Stage 6: Tighten an additional 90 degrees.
    - Stage 7: Tighten an additional 90 degrees.



41. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraper to clean the sealing surfaces.**

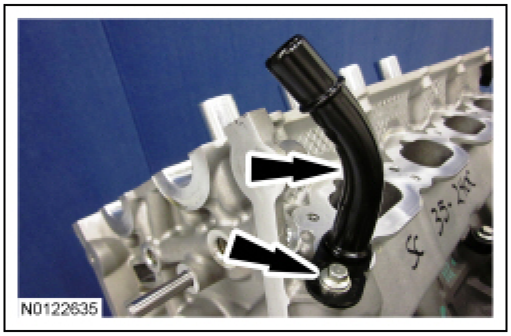
    Clean the exhaust manifold mating surface of the cylinder head with metal surface prep. Follow the directions on the packaging.

42. Install the Cylinder Head Temperature (CHT) sensor.
    - Tighten to 11 Nm (97 lb-in).

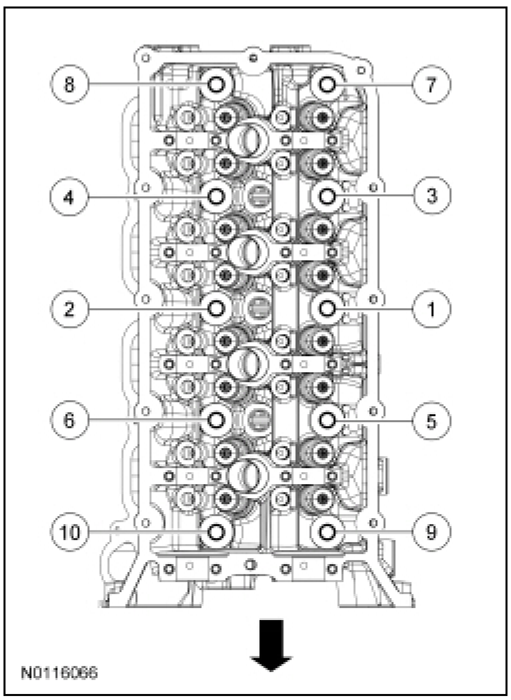


43. **NOTICE: Lubricate the O-ring seal with clean engine coolant prior to installing.**

    Using a new O-ring seal, install the coolant outlet pipe.
    - Tighten to 10 Nm (89 lb-in).



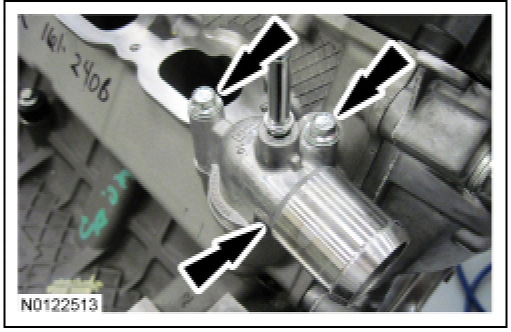
44. Tighten the new LH cylinder head bolts in 7 stages in the sequence shown.
    - Stage 1: Tighten to 25 Nm (18 lb-ft).
    - Stage 2: Tighten to 75 Nm (55 lb-ft).
    - Stage 3: **Loosen** 180 degrees.
    - Stage 4: Tighten to 25 Nm (18 lb-ft).
    - Stage 5: Tighten to 40 Nm (30 lb-ft).
    - Stage 6: Tighten an additional 90 degrees.
    - Stage 7: Tighten an additional 90 degrees.



45. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraper to clean the sealing surfaces.**

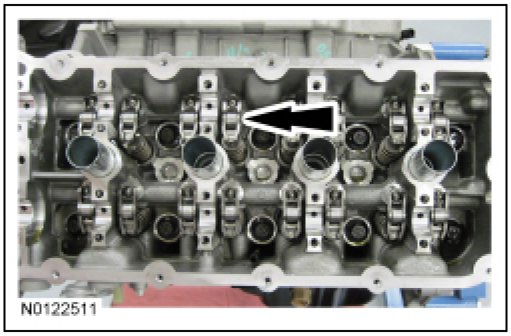
    Clean the exhaust manifold mating surface of the cylinder head with metal surface prep. Follow the directions on the packaging.

46. Using a new gasket, install the coolant outlet and 2 bolts.
    - Tighten to 10 Nm (89 lb-in).



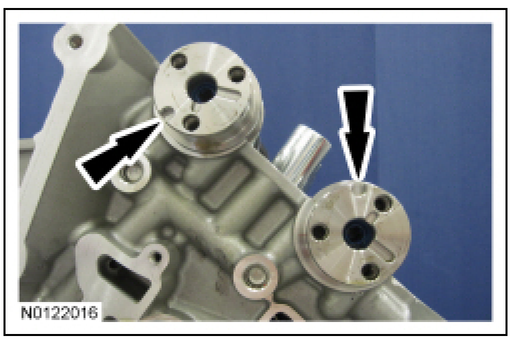
47. **NOTE:** *Lubricate the camshaft roller follower and hydraulic lash adjuster assemblies with clean engine oil prior to installation.*

    Install the 16 camshaft roller follower and hydraulic lash adjuster assemblies.

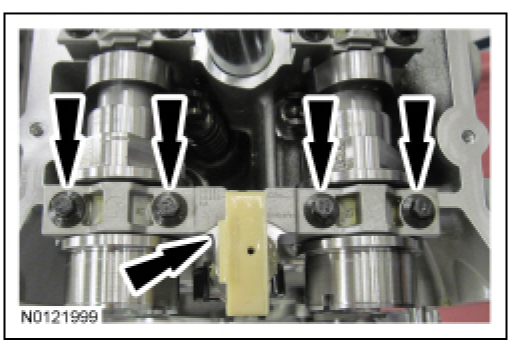


48. **NOTE:** *Lubricate the camshafts with clean engine oil prior to installation.*

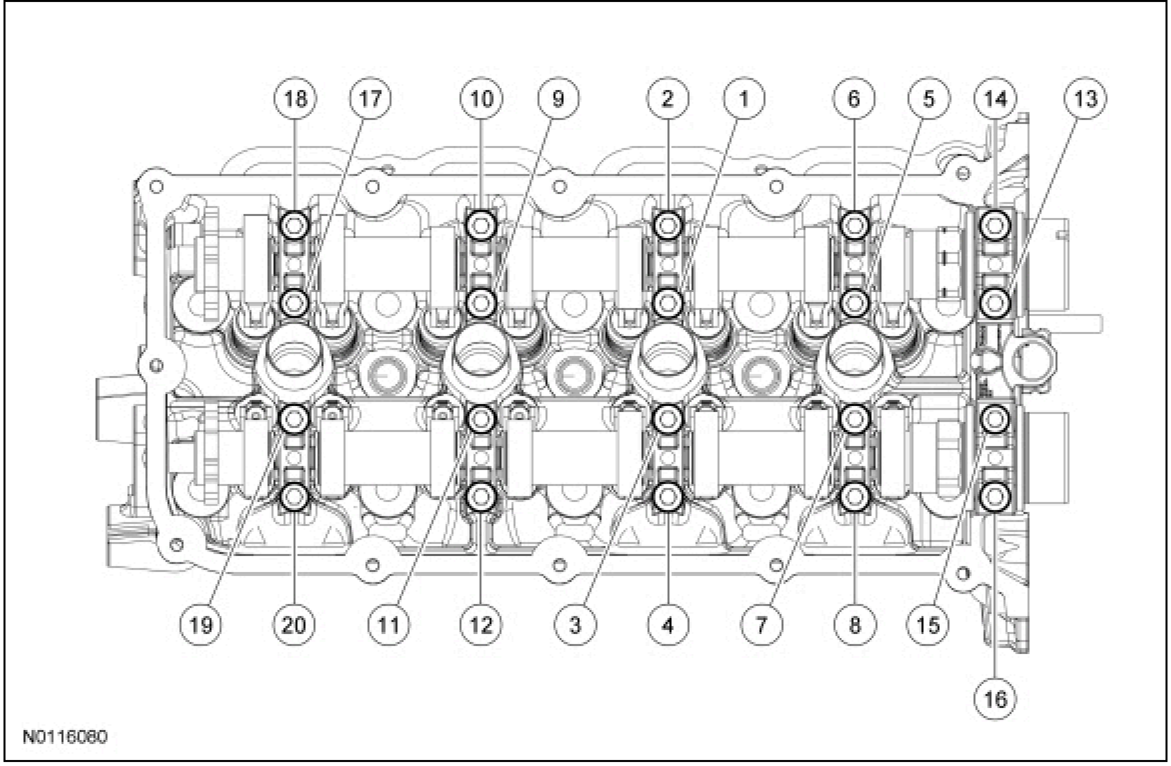
    Install the LH intake and exhaust camshafts in the neutral position. Align the D-slots as shown in the illustration.



49. Install the 8 camshaft bearing caps and the 16 bolts. Do not tighten the bolts at this time.
50. Install the LH front camshaft bearing mega cap and the 4 bolts. Do not tighten the bolts at this time

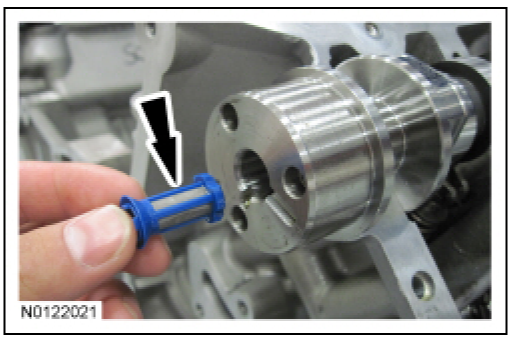


51. Tighten the bolts in the sequence shown in 2 stages.
    - Stage 1: Tighten to 6 Nm (53 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

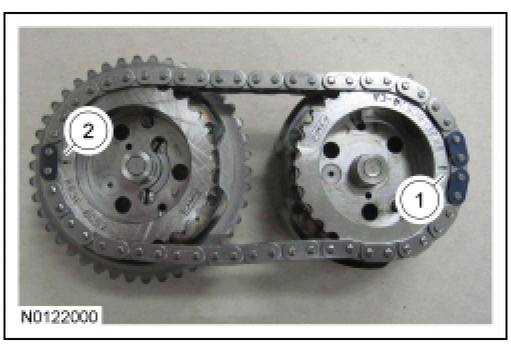


52. **NOTE:** *Intake camshaft shown, exhaust camshaft similar.*

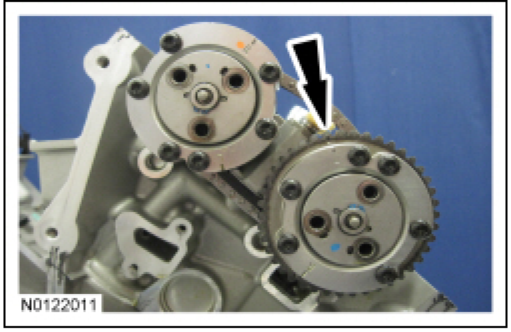
    Install the Variable Camshaft Timing (VCT) system oil filter in the intake and exhaust camshafts.



53. Install the secondary timing chain onto the LH VCT assemblies. Align the colored links on the secondary timing chain with the timing marks on the VCT assemblies as shown in the illustration.
    1. The timing mark on the intake VCT assembly should align between the 2 consecutive colored links.
    2. The timing mark on the exhaust VCT assembly should align with the single colored link.

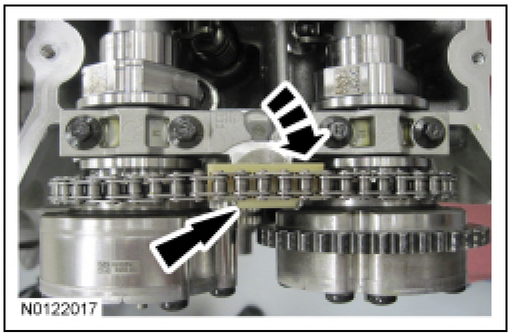


54. Install the LH VCT assemblies and the secondary timing chain onto the LH camshafts to a position 2 mm (0.078 in) from fully seated. The timing mark on the exhaust VCT assembly should be in the 11 o'clock position.



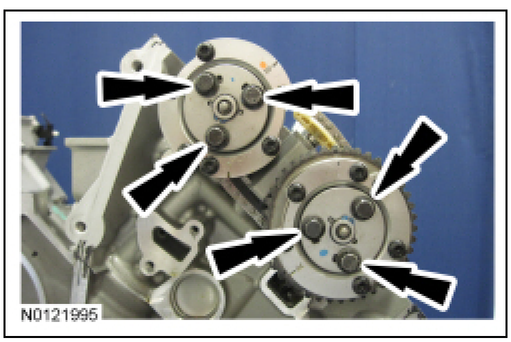
55. **NOTE:** *It may be necessary to rotate the exhaust camshaft slightly (using a wrench on the flats of the camshaft) to seat the VCT assemblies onto the camshafts.*

    Rotate the secondary timing chain tensioner 90 degrees so the ramped area is facing forward and fully seat the VCT assemblies onto the camshafts.
    - If the secondary timing chain is not centered over the tensioner, reposition the VCT assemblies until they are fully seated on the camshafts.

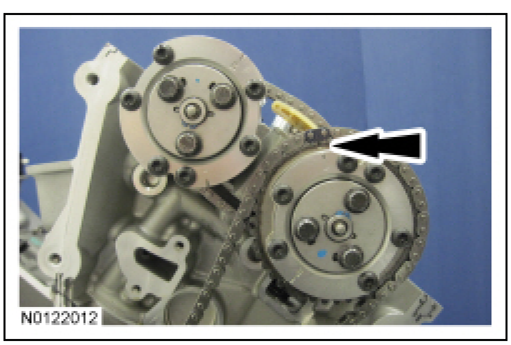


56. **NOTE:** *Use a wrench on the flats of the camshaft to hold the camshafts while tightening the VCT assembly bolts.*

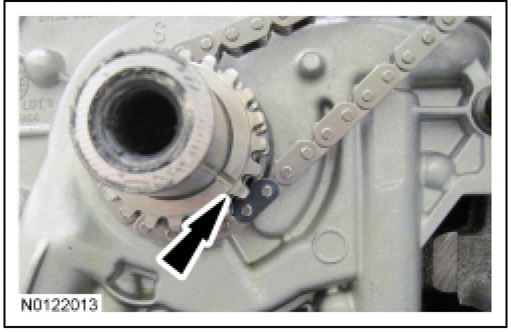
    Install the 3 new LH intake VCT assembly bolts and the 3 new LH exhaust VCT assembly bolts.
    - Tighten to 15 Nm (133 lb-in) plus an additional 90 degrees.



57. Install the LH primary timing chain.
    - Align the colored link on the timing chain with the timing mark on the LH VCT assembly.

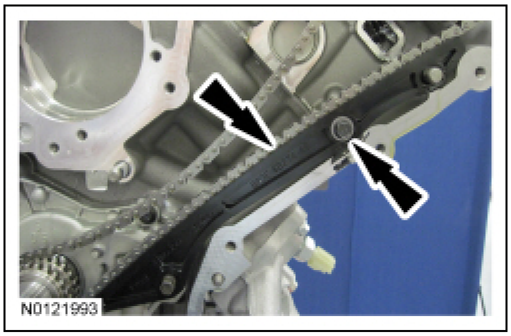


58. Align the remaining colored link on the timing chain with the timing mark on the crankshaft sprocket.



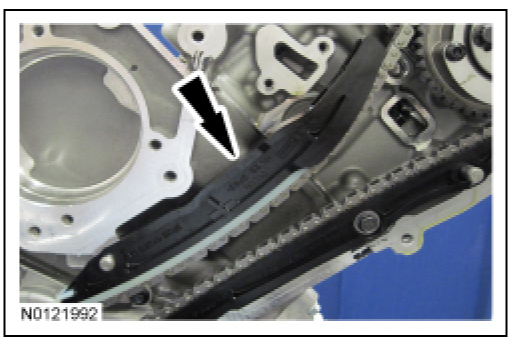
59. **NOTE:** *It may be necessary to rotate the crankshaft slightly to provide enough slack in the chain to install the LH timing chain guide. Return the crankshaft keyway to the 9 o'clock position after installing the LH timing chain guide.*

    Install the LH timing chain guide and bolt.
    - Tighten to 10 Nm (89 lb-in).



60. **NOTE:** *It may be necessary to rotate the crankshaft slightly to provide enough slack in the chain to install the LH timing chain tensioner arm. Return the crankshaft keyway to the 9 o'clock position after installing the LH timing chain tensioner arm.*

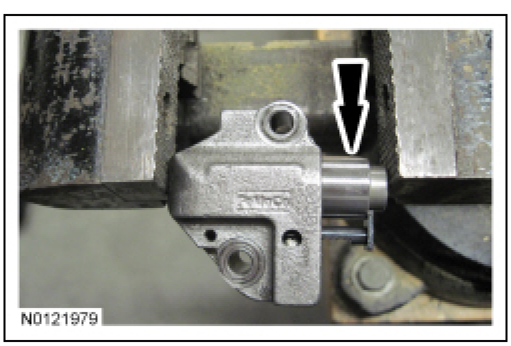
    Install the LH timing chain tensioner arm.



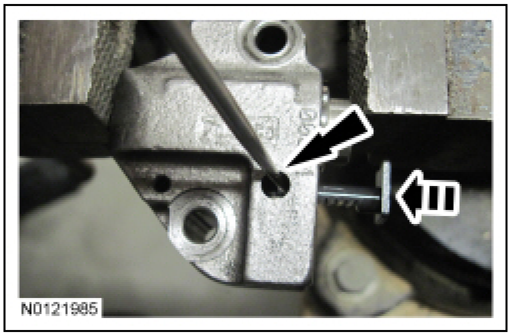
NOTE: *Complete the following 3 steps on both the LH and RH primary timing chain tensioners.*

61. **NOTICE: Do not compress the ratchet assembly or damage to the tensioner will occur.**

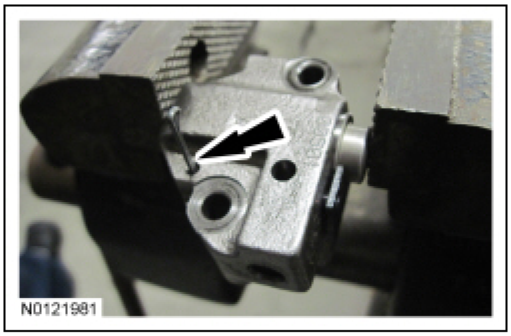
    Compress the primary timing chain tensioner plunger, using an edge of a vise.



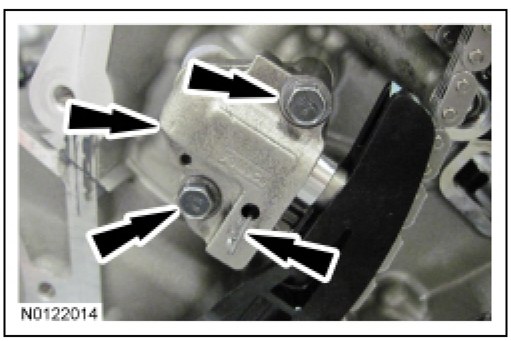
62. Using a small screwdriver or pick, push back and hold the ratchet mechanism, then push the ratchet arm back into the tensioner housing.



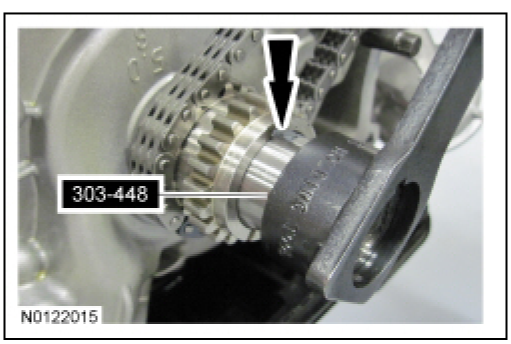
63. Install a suitable pin into the hole of the tensioner housing to hold the ratchet assembly and plunger in place during installation.



64. Install the LH primary timing chain tensioner and 2 bolts.
    - Tighten to 10 Nm (89 lb-in).
    - Remove the holding pin from the tensioner.

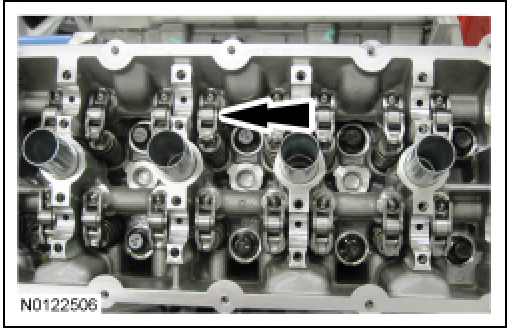


65. Using the crankshaft holding tool, rotate the crankshaft clockwise until the crankshaft keyway is at the 12 o'clock position.



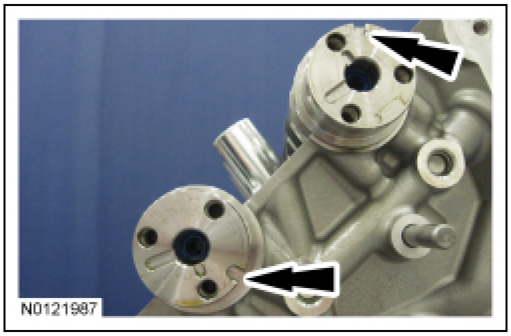
66. **NOTE:** *Lubricate the camshaft roller follower and hydraulic lash adjuster assemblies with clean engine oil prior to installation.*

    Install the 16 camshaft roller follower and hydraulic lash adjuster assemblies.

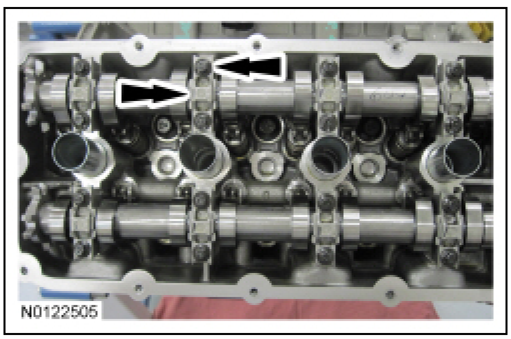


67. **NOTE:** *Lubricate the camshafts with clean engine oil prior to installation.*

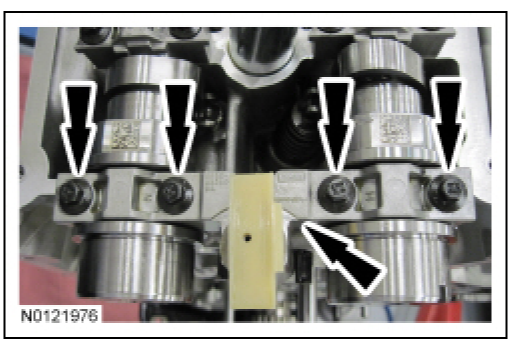
Install the RH intake and exhaust camshafts in the neutral position. Align the D-slots as shown in the illustration.



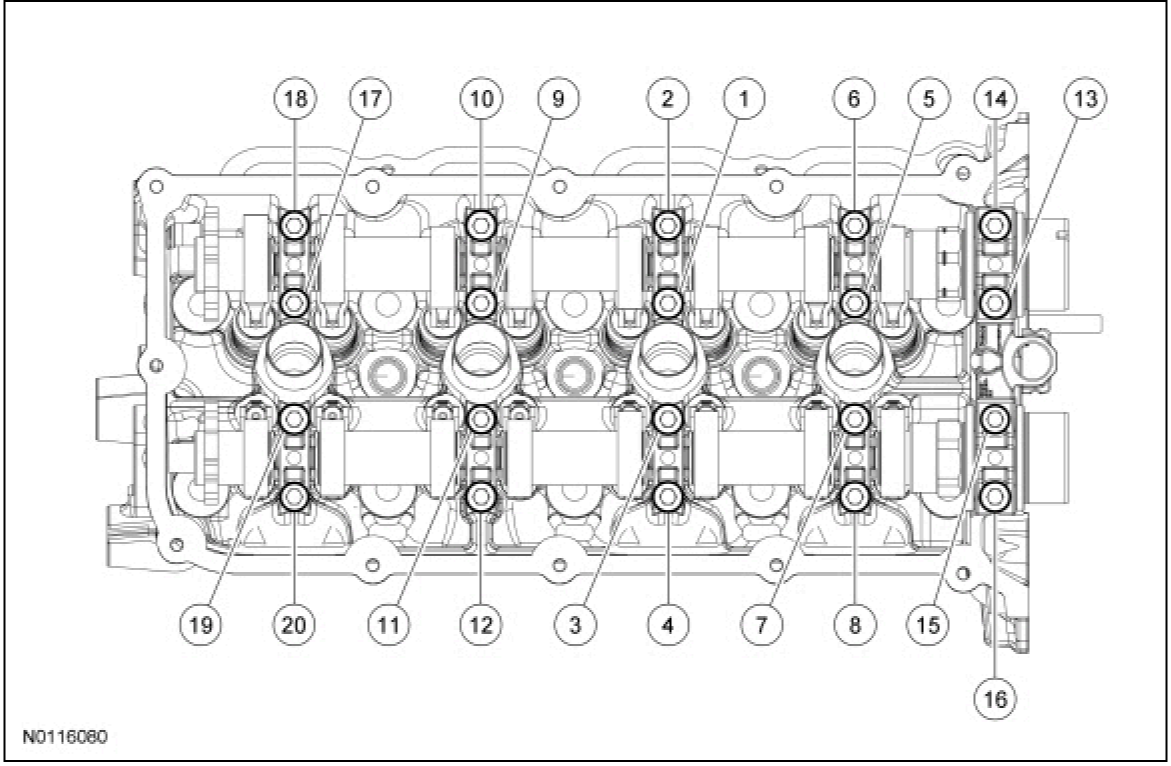
68. Install the 8 camshaft bearing caps and the 16 bolts. Do not tighten the bolts at this time.



69. Install the RH front camshaft bearing mega cap and the 4 bolts. Do not tighten at this time

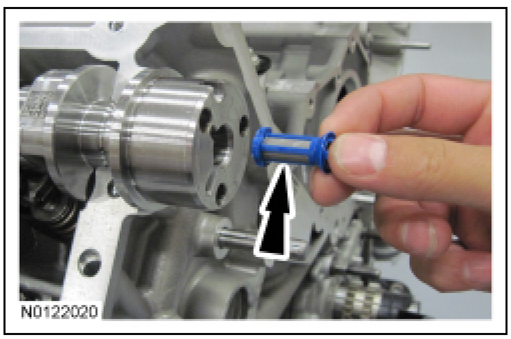


70. Tighten the bolts in the sequence shown in 2 stages.
    - Stage 1: Tighten to 6 Nm (53 lb-in).
    - Stage 2: Tighten an additional 45 degrees.

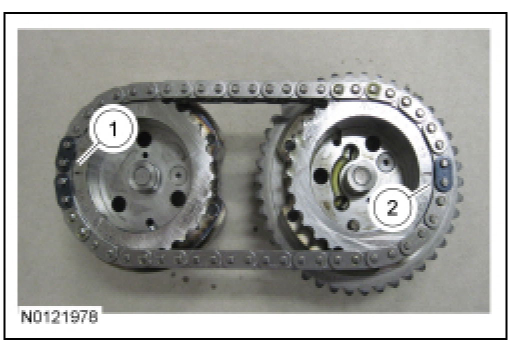


71. **NOTE:** *Intake camshaft shown, exhaust camshaft similar.*

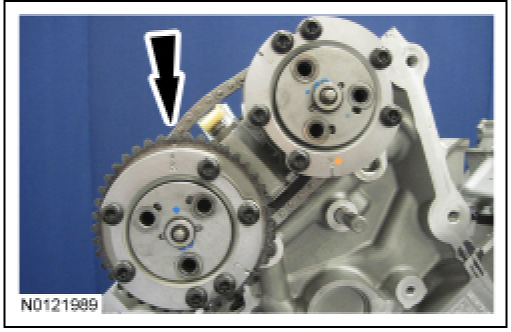
    Install the VCT system oil filter in the intake and exhaust camshafts.



72. Install the secondary timing chain onto the RH VCT assemblies. Align the colored links on the secondary timing chain with the timing marks on the VCT assemblies as shown in the illustration.
    1. The timing mark on the intake VCT assembly should align between the 2 consecutive colored links.
    2. The timing mark on the exhaust VCT assembly should align with the single colored link.

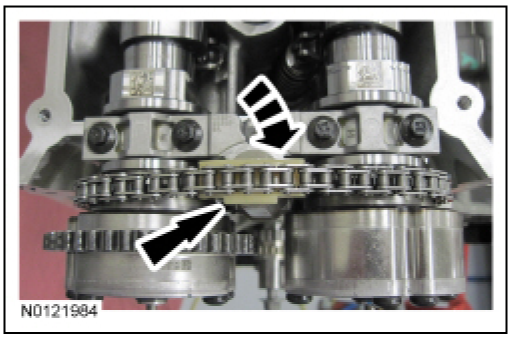


73. Install the RH VCT assemblies and the secondary timing chain onto the RH camshafts to a position 2 mm (0.078 in) from fully seated. The timing mark on the exhaust VCT assembly should be in the 1 o'clock position.



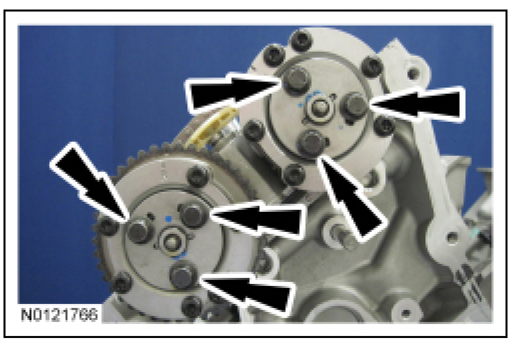
74. **NOTE:** *It may be necessary to rotate the exhaust camshaft slightly (using a wrench on the flats of the camshaft) to seat the VCT assemblies onto the camshafts.*

    Rotate the secondary timing chain tensioner 90 degrees so the ramped area is facing forward and fully seat the VCT assemblies onto the camshafts.
    - If the secondary timing chain is not centered over the tensioner, reposition the VCT assemblies until they are fully seated on the camshafts.

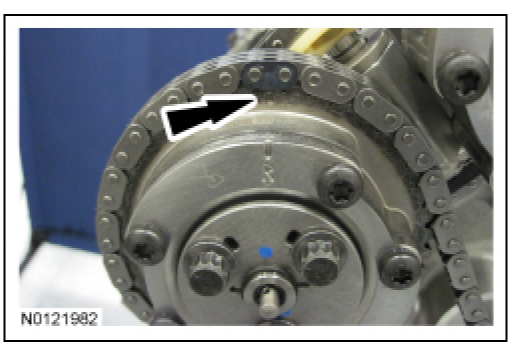


75. **NOTE:** *Use a wrench on the flats of the camshaft to hold the camshafts while tightening the VCT assembly bolts.*

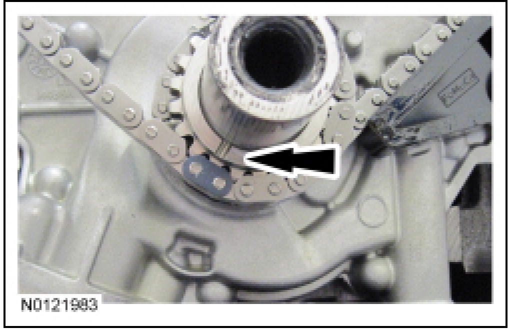
    Install the 3 new RH intake VCT assembly bolts and the 3 new RH exhaust VCT assembly bolts.
    - Tighten to 15 Nm (133 lb-in) plus an additional 90 degrees.



76. Install the RH primary timing chain.
    - Align the colored link on the timing chain with the timing mark on the RH VCT assembly.

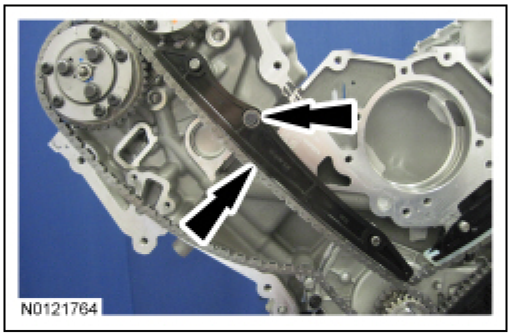


77. Align the remaining colored link on the timing chain with the timing mark on the crankshaft sprocket.



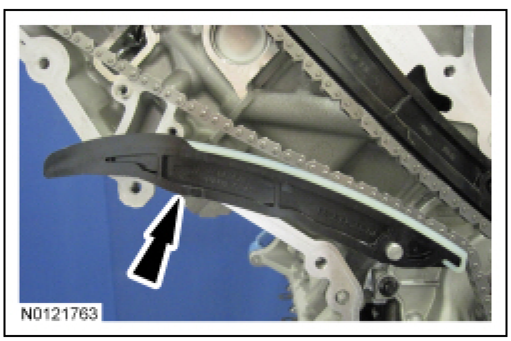
78. **NOTE:** *It may be necessary to rotate the crankshaft slightly to provide enough slack in the chain to install the RH timing chain guide. Return the crankshaft keyway to the 12 o'clock position after installing the RH timing chain guide.*

    Install the RH timing chain guide and bolt.
    - Tighten to 10 Nm (89 lb-in).

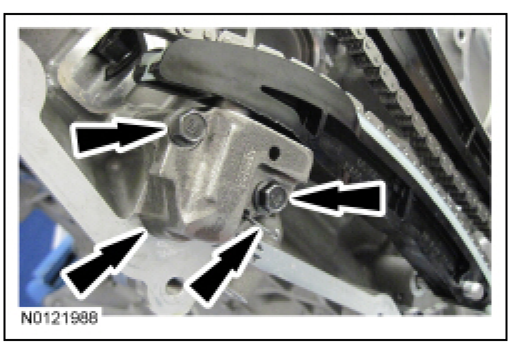


79. **NOTE:** *It may be necessary to rotate the crankshaft slightly to provide enough slack in the chain to install the RH timing chain tensioner arm. Return the crankshaft keyway to the 12 o'clock position after installing the RH timing chain tensioner arm.*

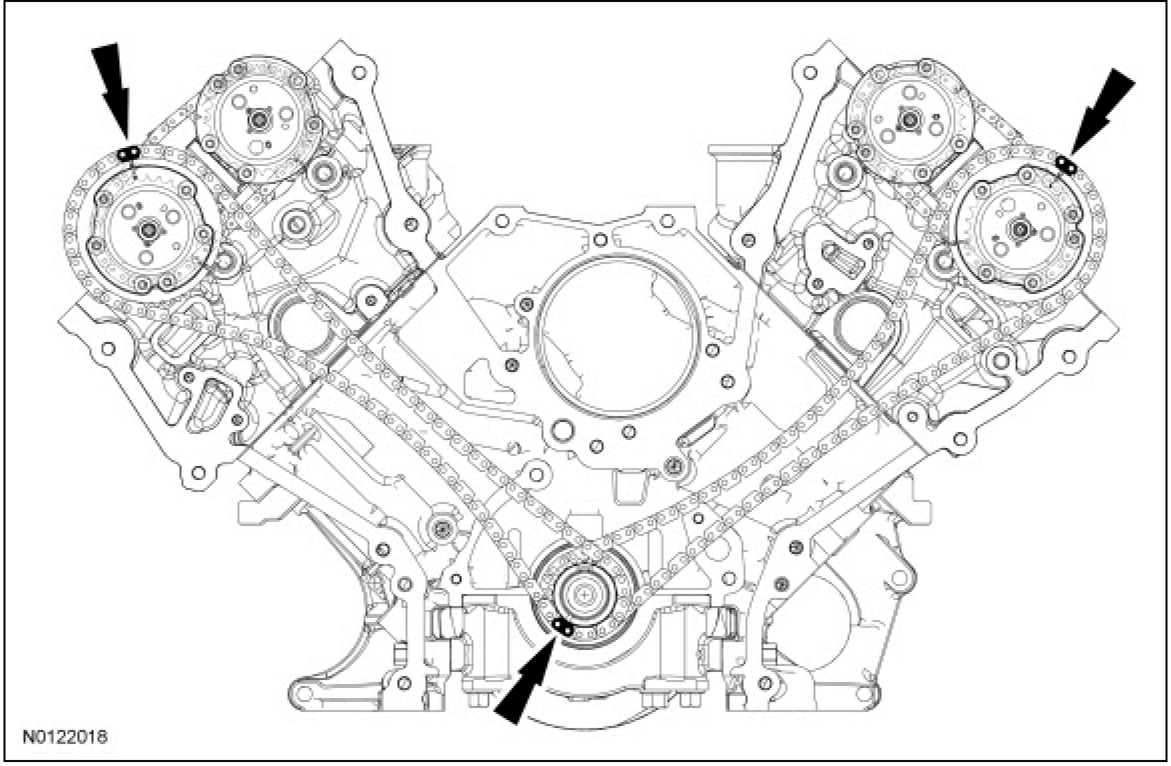
    Install the RH timing chain tensioner arm.



80. Install the RH primary timing chain tensioner and 2 bolts.
    - Tighten to 10 Nm (89 lb-in).
    - Remove the holding pin from the tensioner.



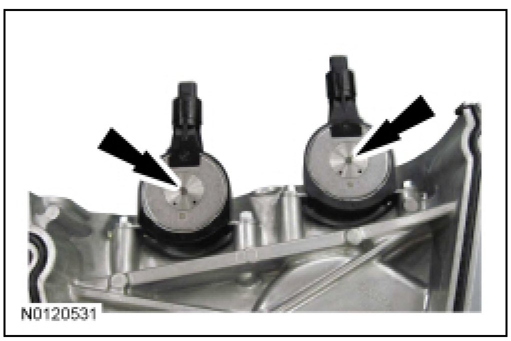
81. With the crankshaft keyway still at the 12 o'clock position, verify the timing mark alignment is correct.



82. **NOTICE: The Variable Camshaft Timing (VCT) variable force solenoid pins must be fully depressed to avoid interference with the VCT valve tips when installing the engine front cover. Failure to follow these instructions can result in damage to the engine.**

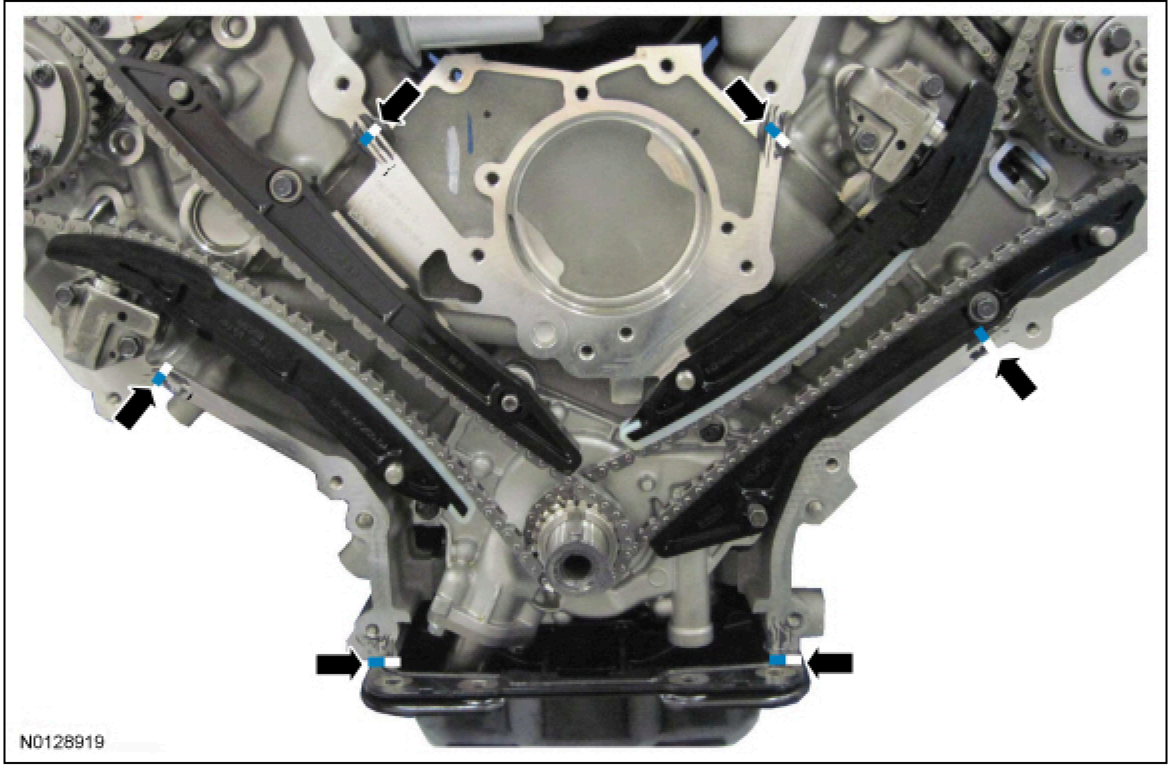
    NOTE: *LH shown, RH similar.*

    Fully depress the VCT variable force solenoid pins.



83. **NOTE:** *The engine front cover must be installed and all fasteners final tightened within 5 minutes of applying the sealer. If this cannot be accomplished, install the engine front cover and tighten fasteners 6, 7, 8, 9, 10 and 11 to 8 Nm (71 lb-in) within 5 minutes of applying the sealer. All of the fasteners must then be final tightened within 1 hour of applying the sealer. If this time limit is exceeded, all sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

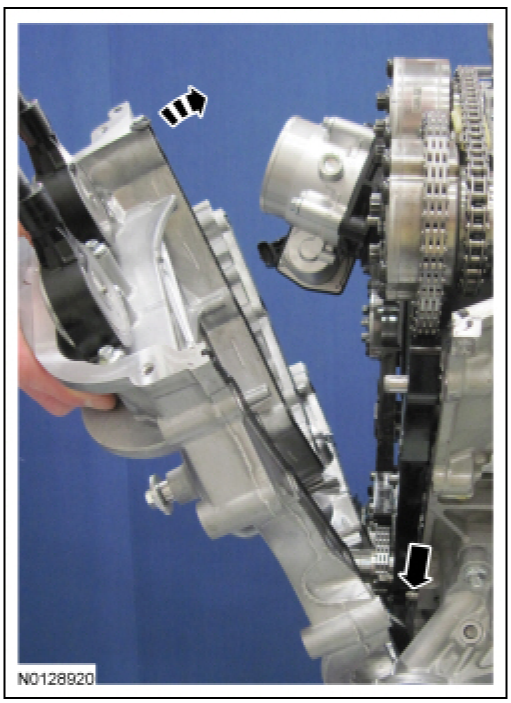
    Apply a bead of silicone gasket and sealant to the cylinder head-to-cylinder block joints and the oil pan-to-cylinder block joints as illustrated.



84. **NOTE:** *Make sure that the engine front cover gasket is in place on the engine front cover before installation.*

Using new gaskets, position the engine front cover onto the dowels.

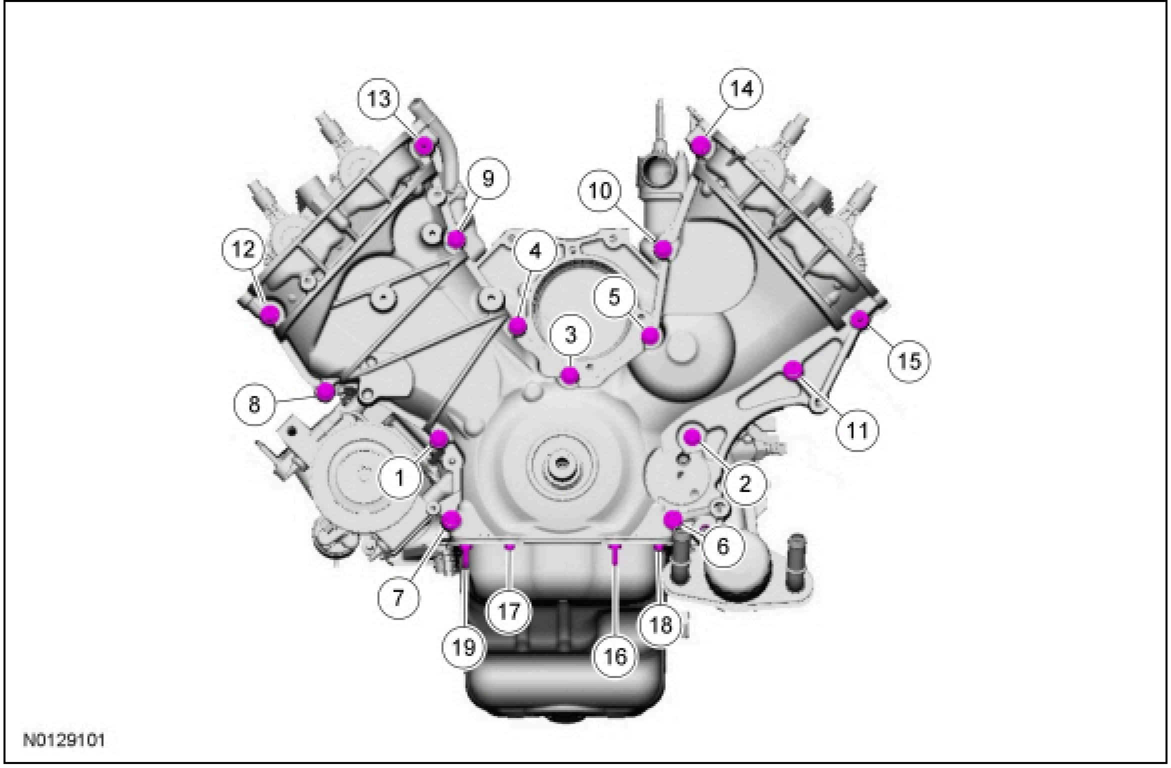
- Install the 13 engine front cover bolts, 2 stud bolts, 2 oil pan-to-engine front cover bolts and 2 oil pan-to-engine front cover stud bolts finger-tight.



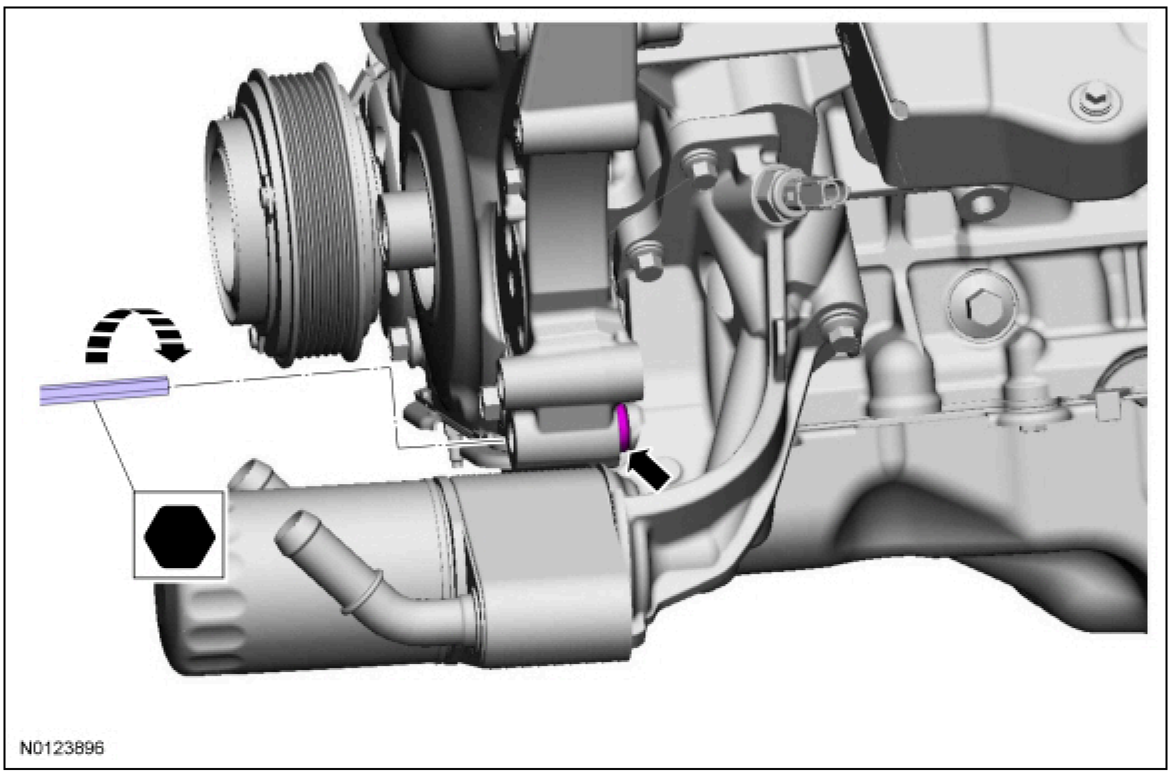
85. **NOTE:** *The engine front cover must be installed and all fasteners final tightened within 5 minutes of applying the sealer. If this cannot be accomplished, install the engine front cover and tighten fasteners 6, 7, 8, 9, 10 and 11 to 8 Nm (71 lb-in) within 5 minutes of applying the sealer. All of the fasteners must then be final tightened within 1 hour of applying the sealer. If this time limit is exceeded, all sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

Tighten the bolts in the sequence shown in 2 stages.

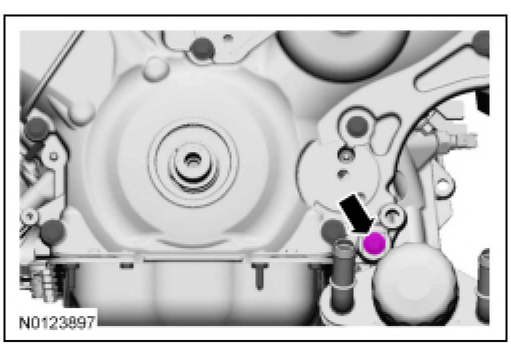
- Stage 1: Tighten bolts 1-15 to 25 Nm (18 lb-ft) and bolts 16-19 to 10 Nm (89 lb-in).
- Stage 2: Tighten bolts 1-15 an additional 60 degrees and bolts 16-19 an additional 45 degrees.



86. Using a 10 mm Hex Bit, rotate the engine front cover jackscrew into contact with the oil filter adapter to 5 Nm (44 lb-in).

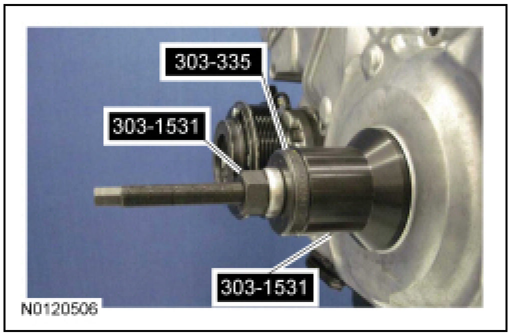


87. Install the engine front cover-to-oil filter adapter bolt and tighten to 25 Nm (18 lb-ft) plus an additional 60 degrees.



88. **NOTE:** *Lubricate the engine front cover bore and the crankshaft front oil seal inner lip with clean engine oil.*

Using the Front Crank Seal and Damper Installer and the Front Cover Oil Seal Installer (plate only), install the crankshaft front oil seal.

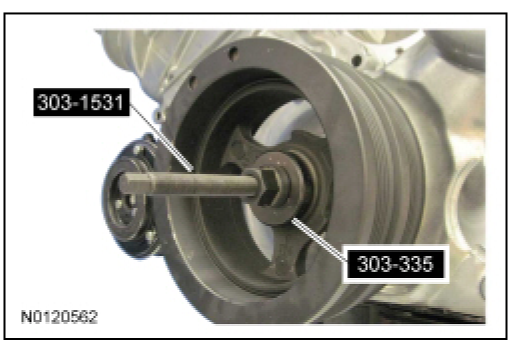


89. **NOTE:** *If not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

Apply silicone gasket and sealant to the Woodruff key slot in the crankshaft pulley.

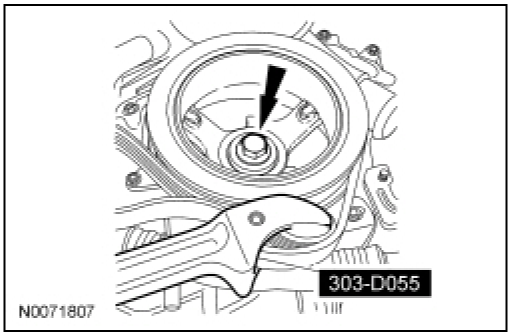
90. Lubricate the crankshaft pulley sealing surface with clean engine oil prior to installation.

91. Using the Front Crank Seal and Damper Installer and the Front Cover Oil Seal Installer (plate only), install the crankshaft pulley.

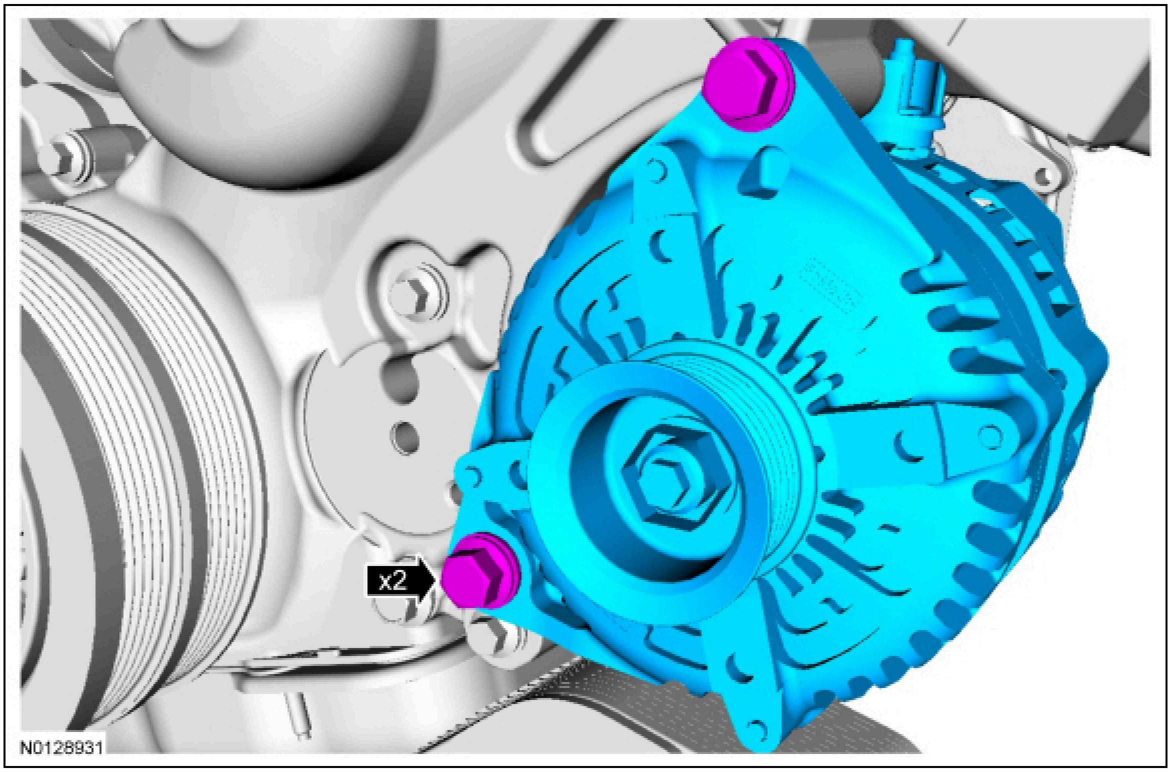


92. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 4 stages:
    - Stage 1: Tighten to 140 Nm (103 lb-ft).
    - Stage 2: Loosen 360 degrees.
    - Stage 3: Tighten to 100 Nm (74 lb-ft).

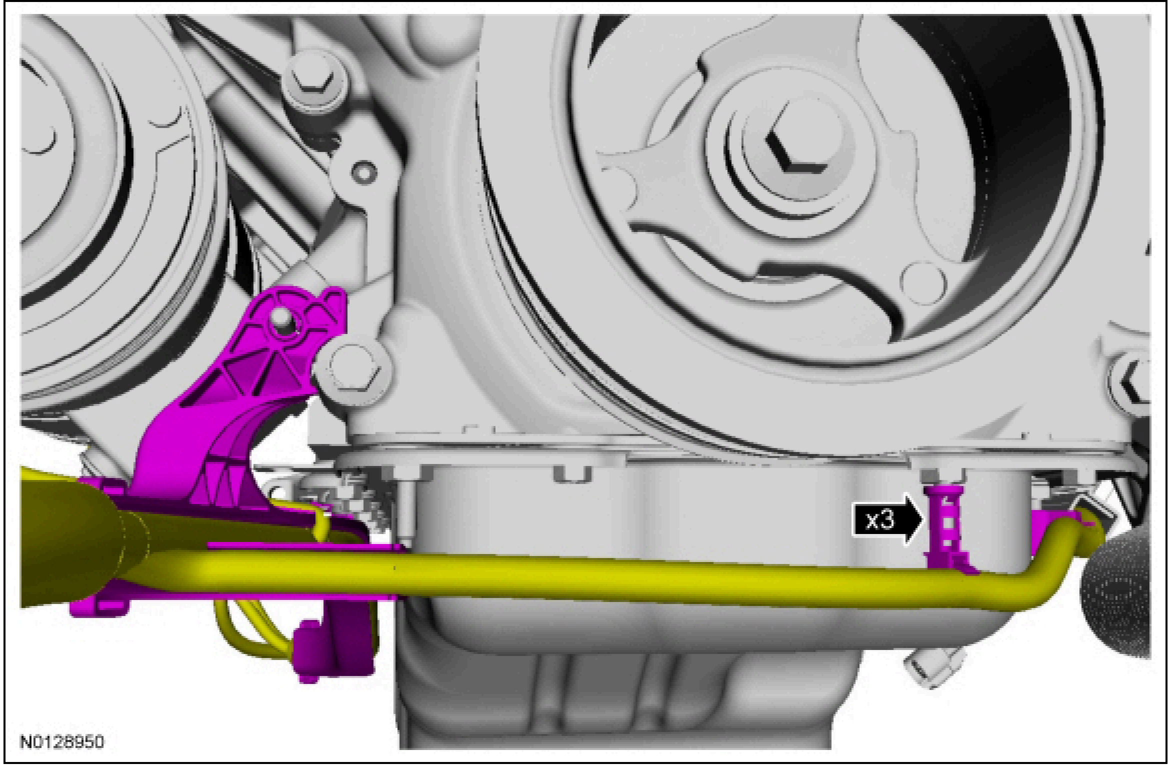
- Stage 4: Tighten an additional 90 degrees.



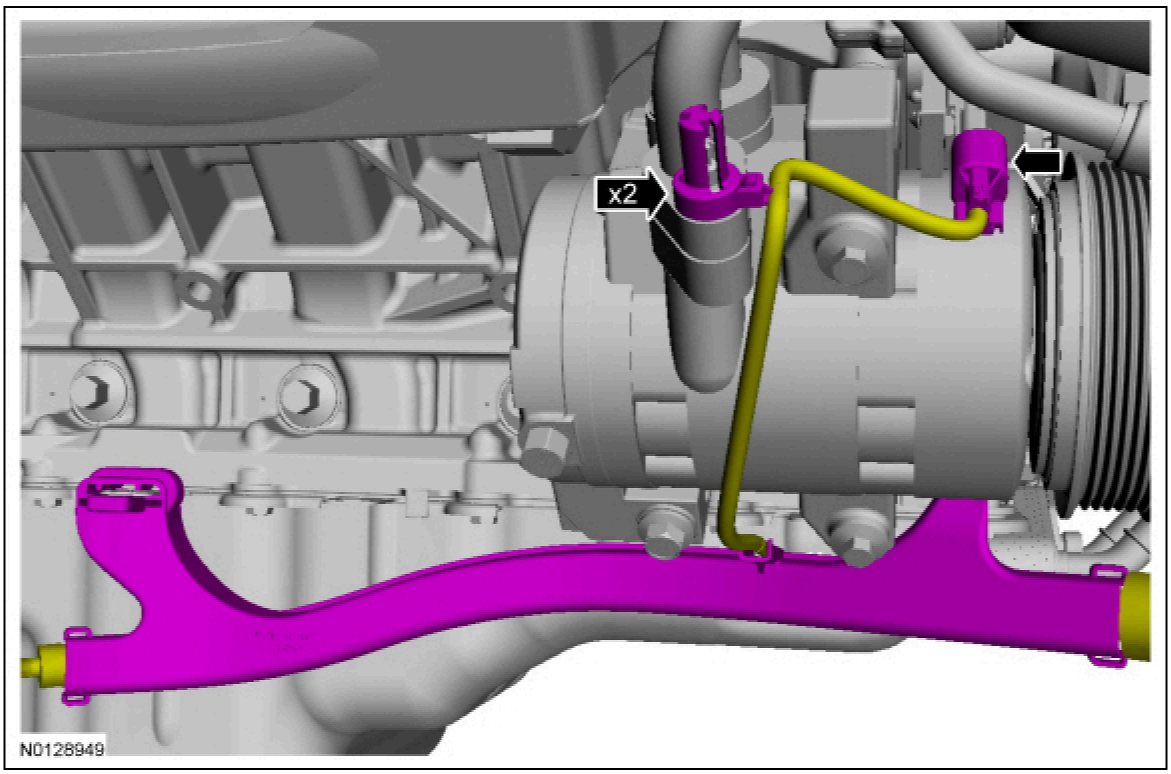
93. Install the generator and the 2 bolts.
    - Tighten to 48 Nm (35 lb-ft).



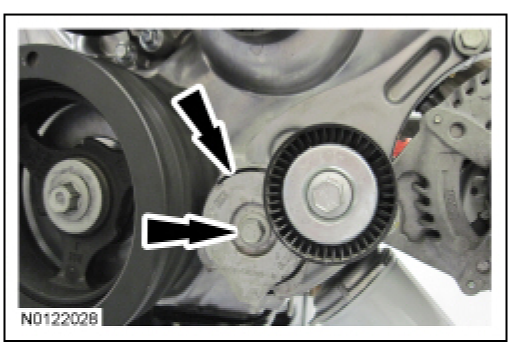
94. Attach the 3 wiring harness retainers.



95. Connect the A/C compressor electrical connector and the 2 wiring harness retainers.



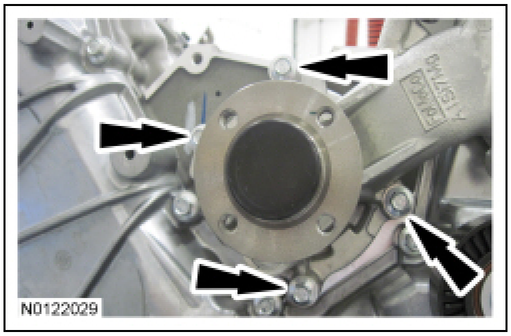
96. Install the accessory drive belt tensioner and the bolt.
    - Tighten to 48 Nm (35 lb-ft).



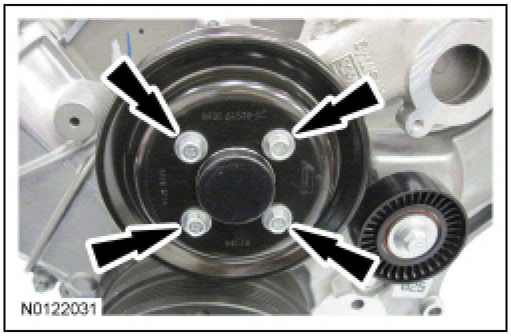
97. **NOTE:** *Lubricate the new coolant pump O-ring seal with clean engine coolant.*

Using a new O-ring seal, install the coolant pump and the 4 bolts.

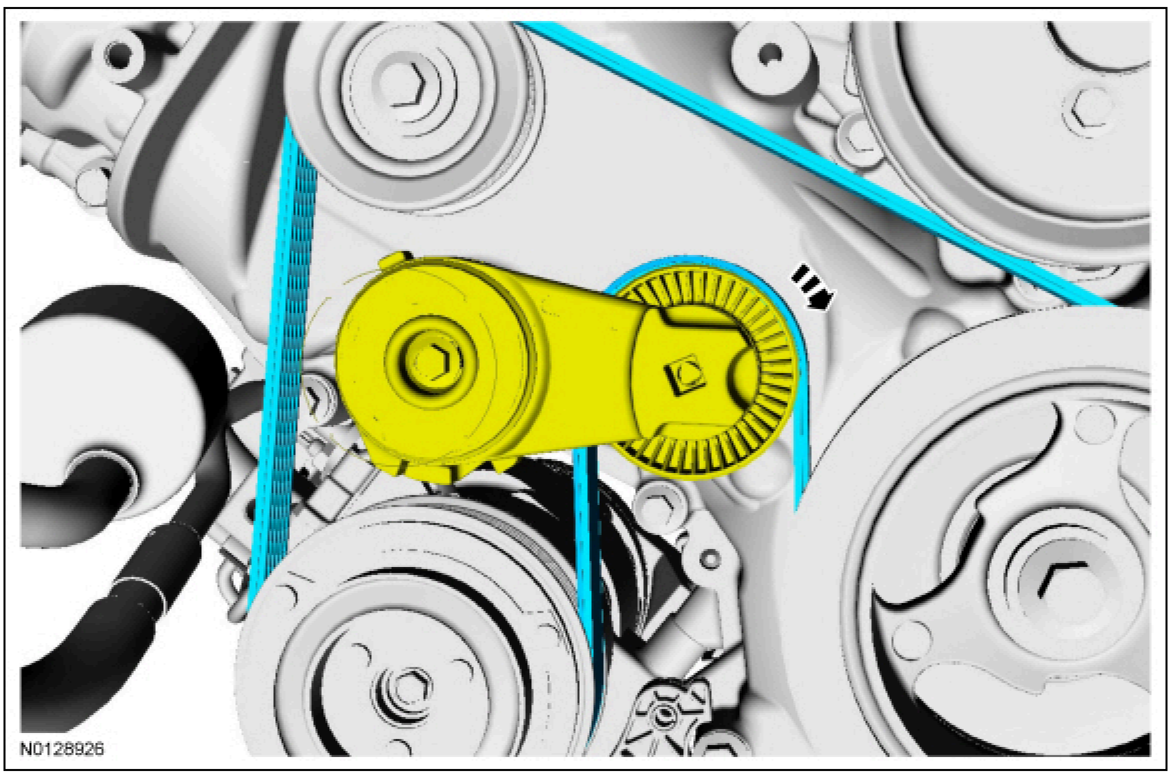
- Tighten to 20 Nm (177 lb-in) plus an additional 60 degrees.



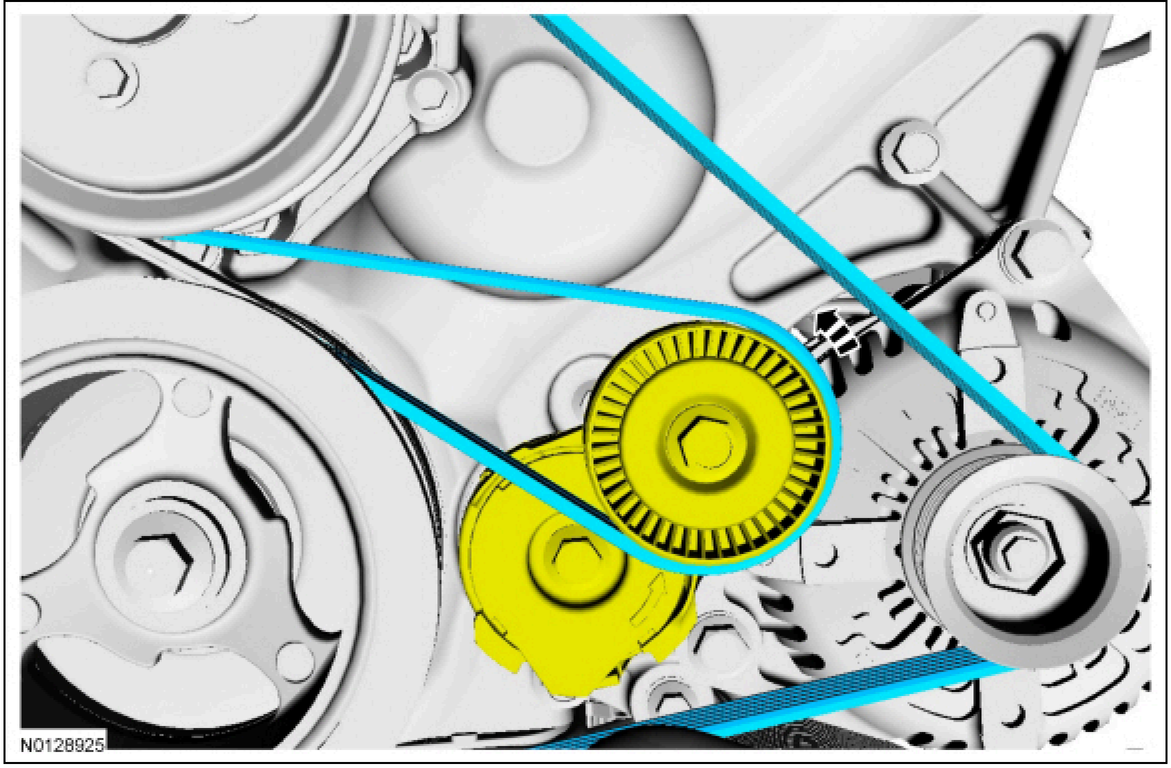
98. Install the coolant pump pulley and the 4 bolts.



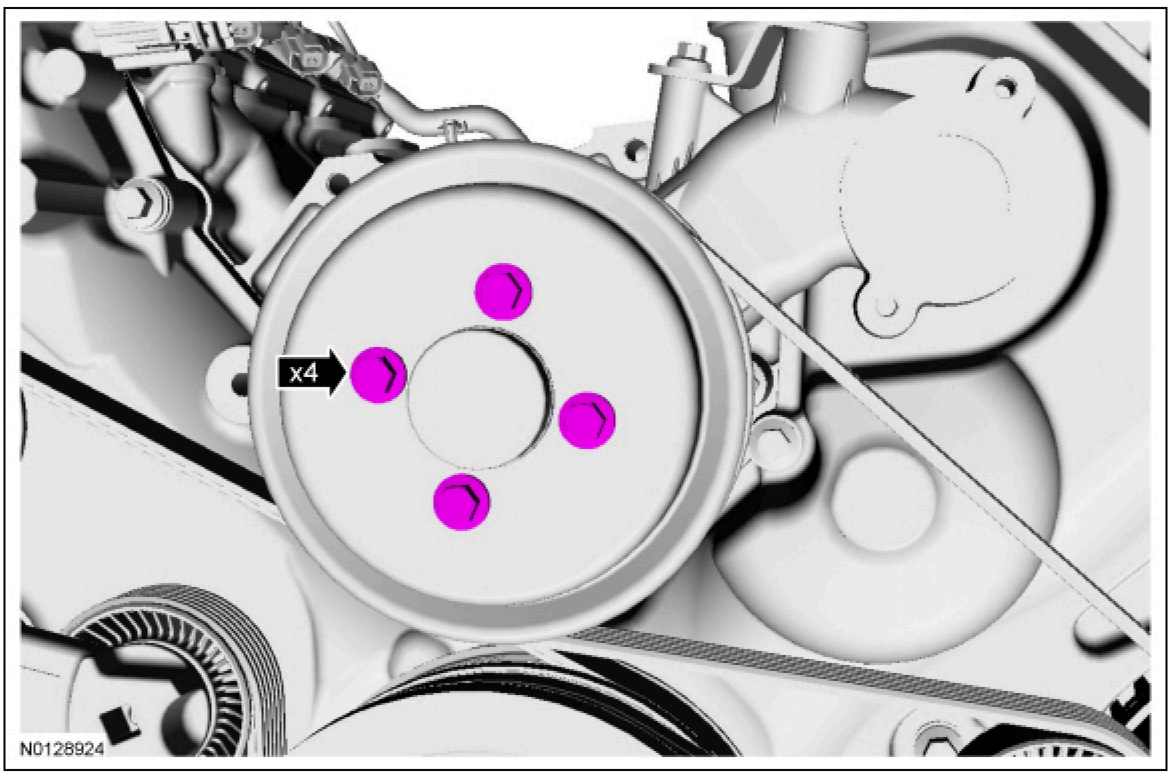
99. Rotate the A/C compressor belt tensioner clockwise and install the A/C compressor belt.



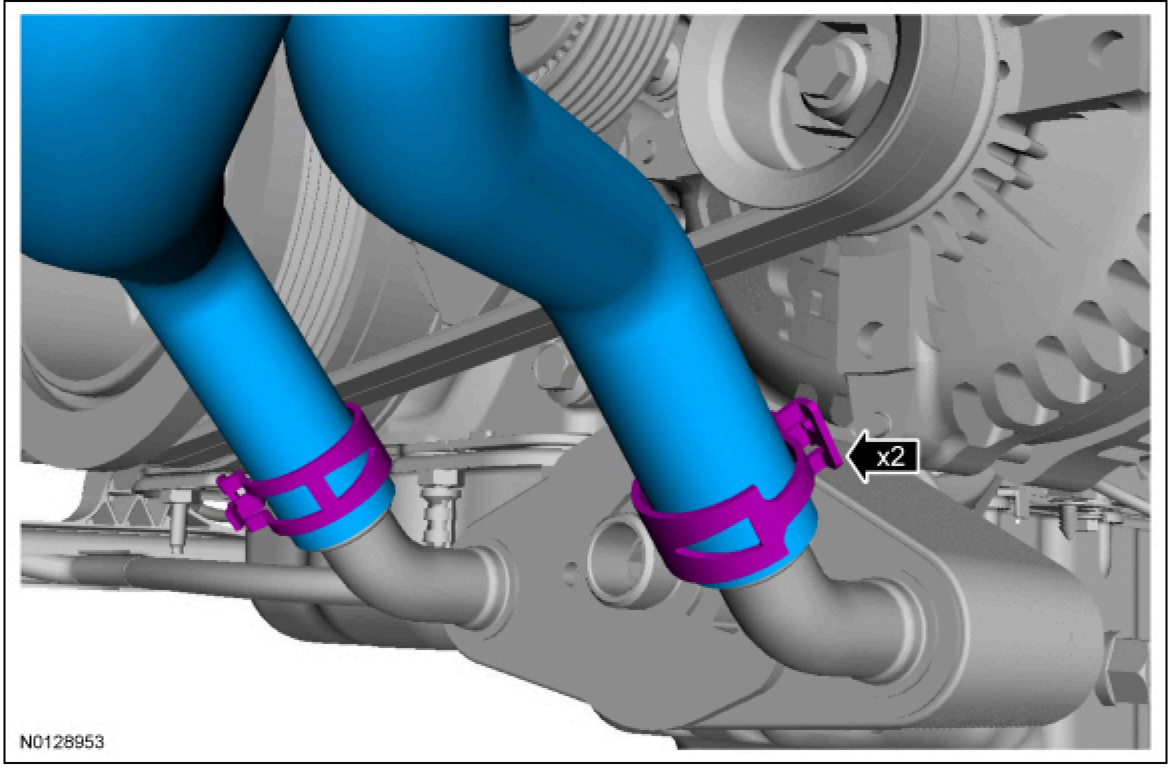
100. Rotate the accessory drive belt tensioner counterclockwise and install the accessory drive belt.



101. Tighten the 4 coolant pump pulley bolts to 25 Nm (18 lb-ft).



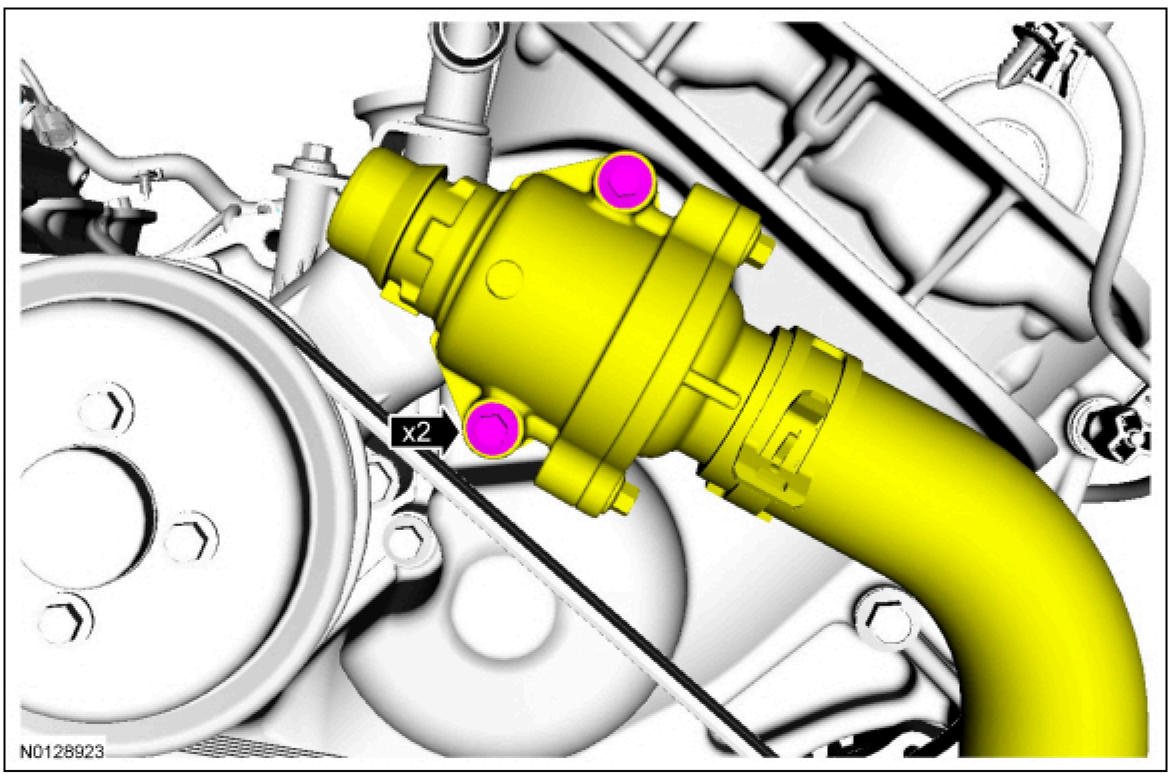
102. Connect the 2 coolant hoses to the oil cooler.



103. **NOTE:** *Lubricate the new thermostat housing O-ring seal with clean engine coolant*.

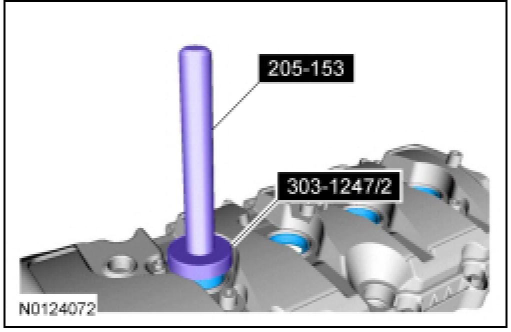
Using a new thermostat housing O-ring seal, install the thermostat housing and the 2 bolts.

- Tighten to 10 Nm (89 lb-in).



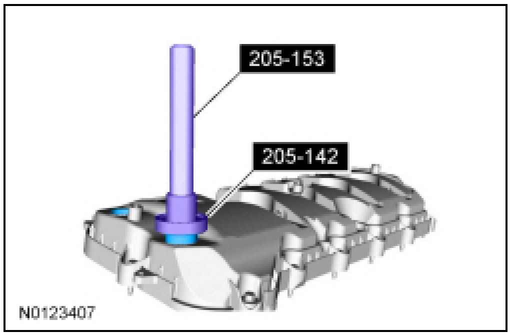
104. **NOTE:** *Installation of new seals is only required if damaged seals were removed*.

Using the VCT Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.



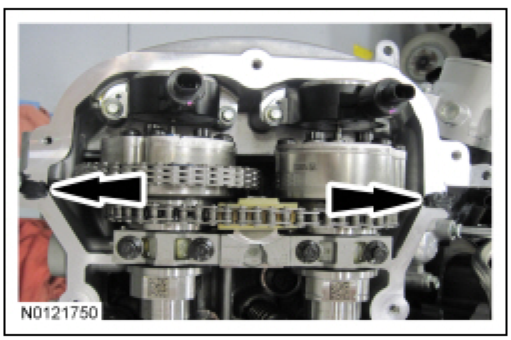
105. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

 Using the Differential Bearing Cone Installer and Handle, install new VCT variable force solenoid seal(s).

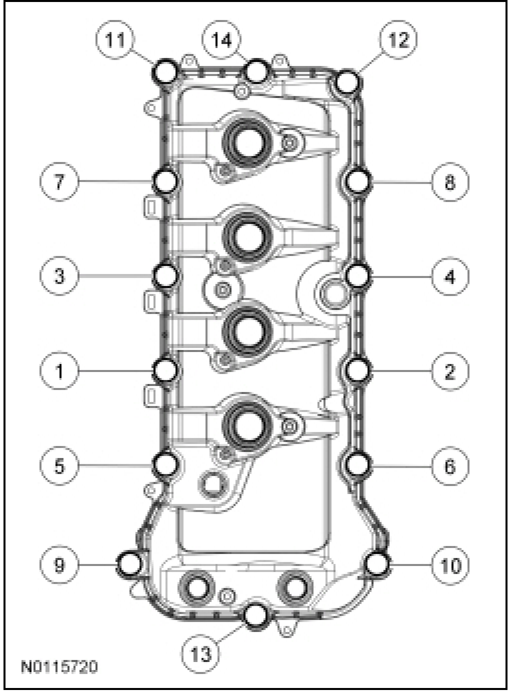


106. **NOTE:** *If the valve cover is not installed and the fasteners tightened within 5 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

 Apply an 8 mm (0.31 in) bead of silicone gasket and sealant to the engine front cover-to-LH cylinder head joints.



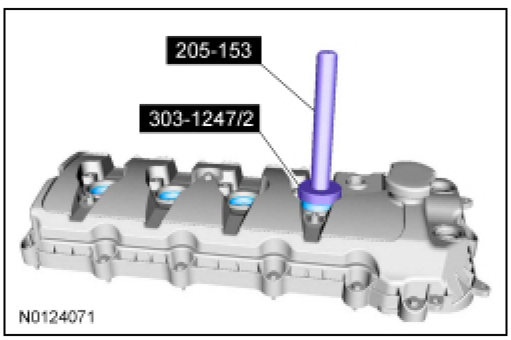
107. Position the LH valve cover and new gasket on the cylinder head.
 - Tighten the bolts in the sequence shown to 10 Nm (89 lb-in).



108. Install the oil level indicator.

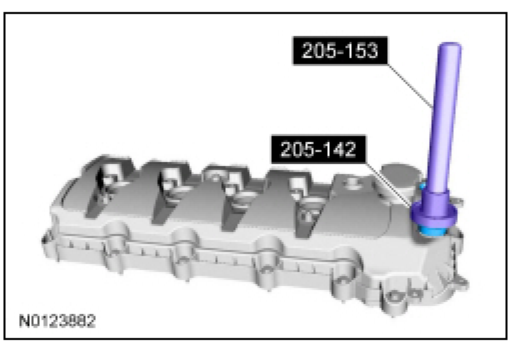
109. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the VCT Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.



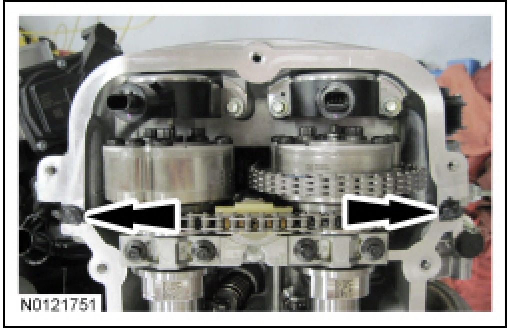
110. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the Differential Bearing Cone Installer and Handle, install new VCT variable force solenoid seal(s).

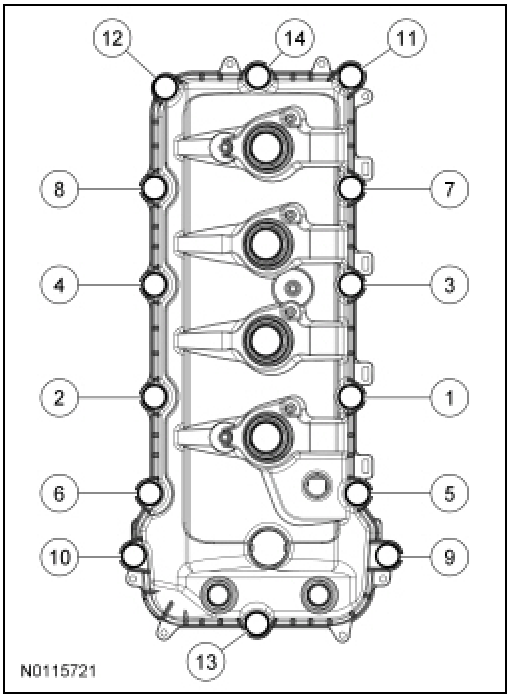


111. **NOTE:** *If the valve cover is not installed and the fasteners tightened within 5 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

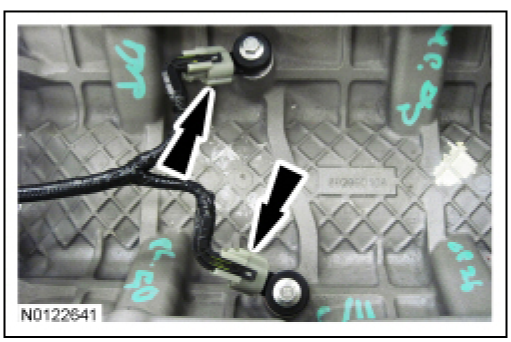
Apply an 8 mm (0.31 in) bead of silicone gasket and sealant to the engine front cover-to-RH cylinder head joints.



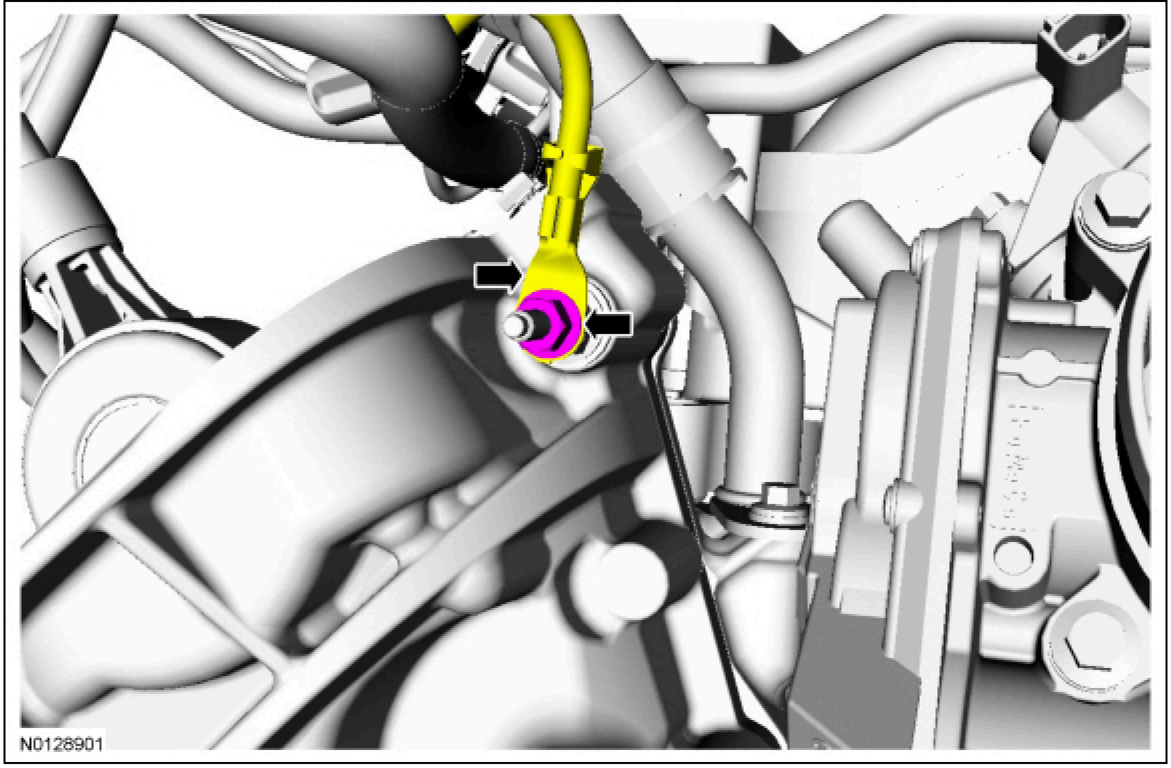
112. Position the RH valve cover and new gasket on the cylinder head.
    - Tighten the bolts in the sequence shown to 10 Nm (89 lb-in).



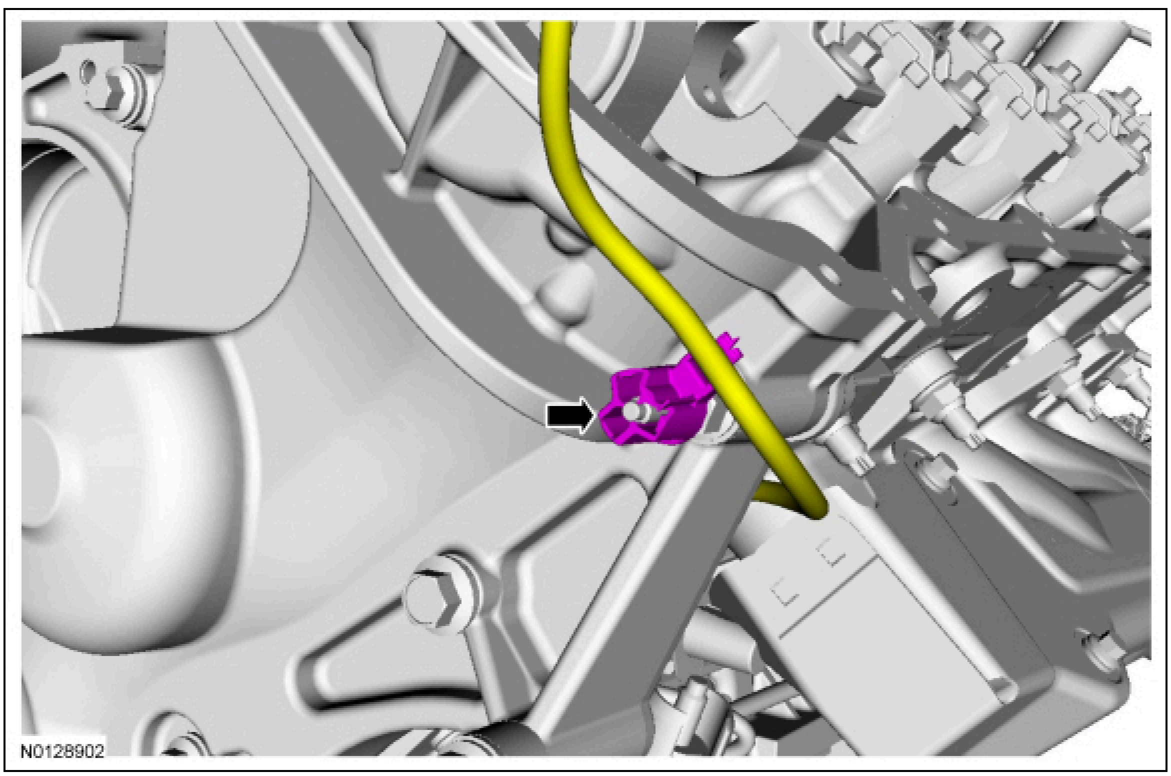
113. Position the wiring harness on the engine.
    - Connect the 2 KS electrical connectors.



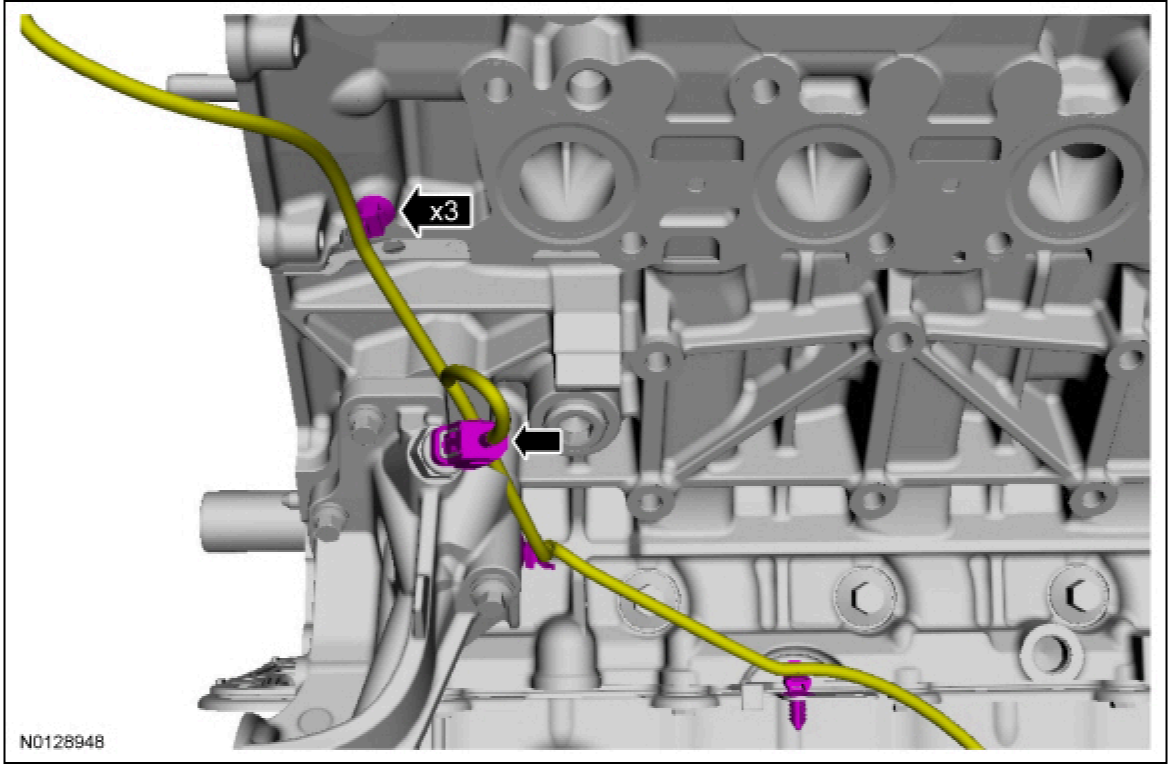
114. Install the ground wire and nut to the engine front cover stud bolt.
    - Tighten to 10 Nm (89 lb-in).



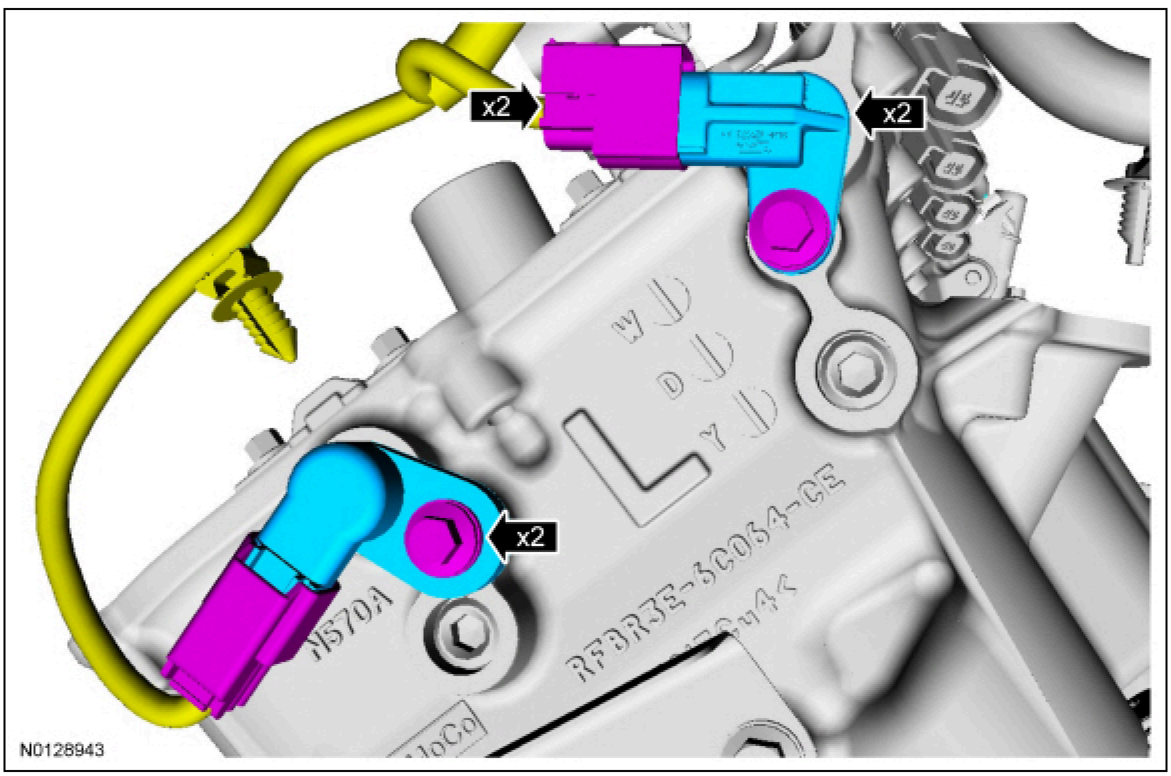
115. Attach the wiring harness retainer to the engine front cover stud bolt.



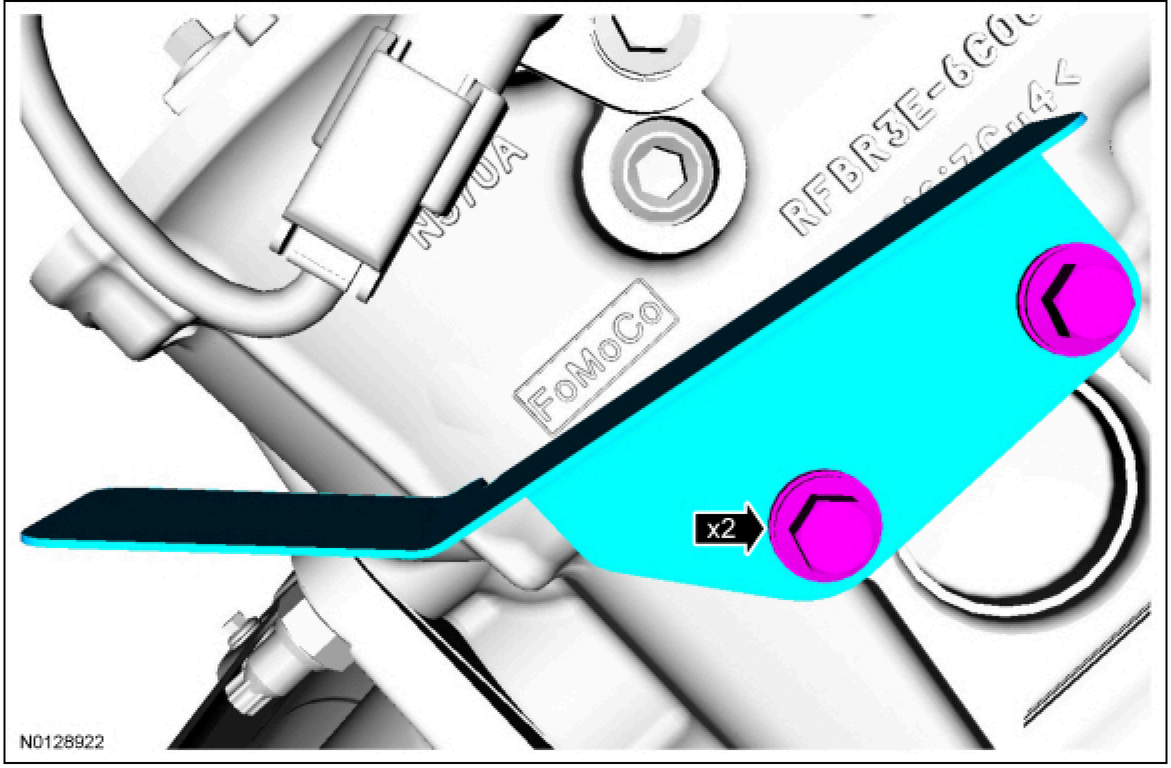
116. Connect the Engine Oil Pressure (EOP) switch electrical connector and the 3 wiring harness retainers.



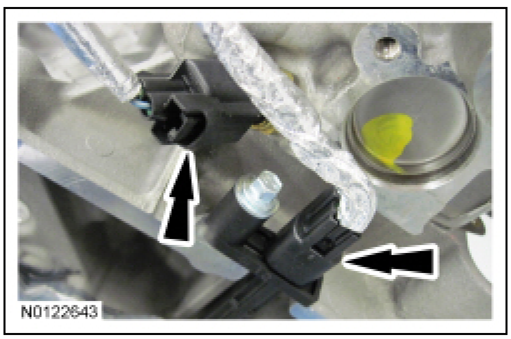
117. If removed, install the LH intake and exhaust Camshaft Position (CMP) sensors and the 2 bolts.
    - Tighten to 10 Nm (89 lb-in).
    - Connect the LH intake and exhaust CMP sensor electrical connectors.



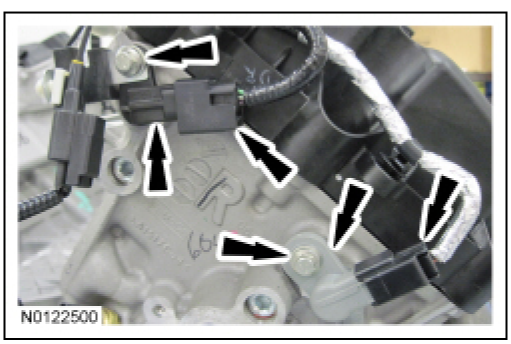
118. Install the heat shield and the 2 bolts to the LH cylinder head.
    - Tighten to 10 Nm (89 lb-in).



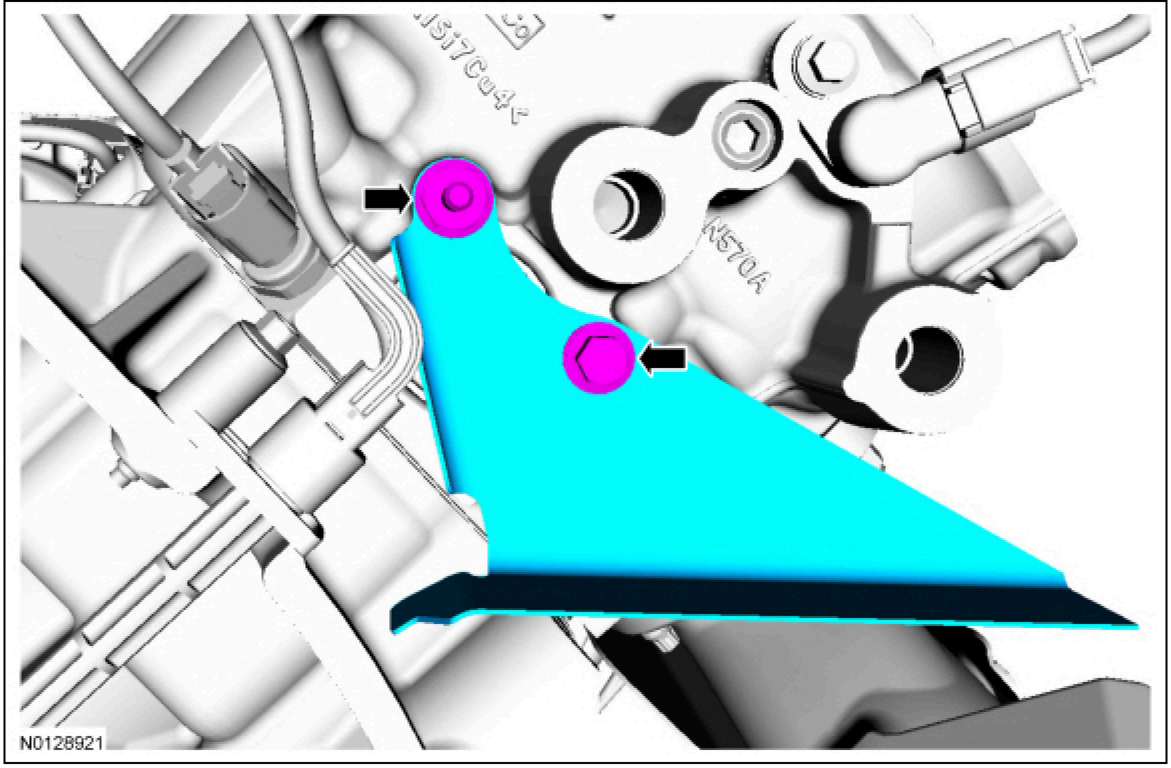
119. Connect the CHT sensor and the CKP sensor electrical connectors.



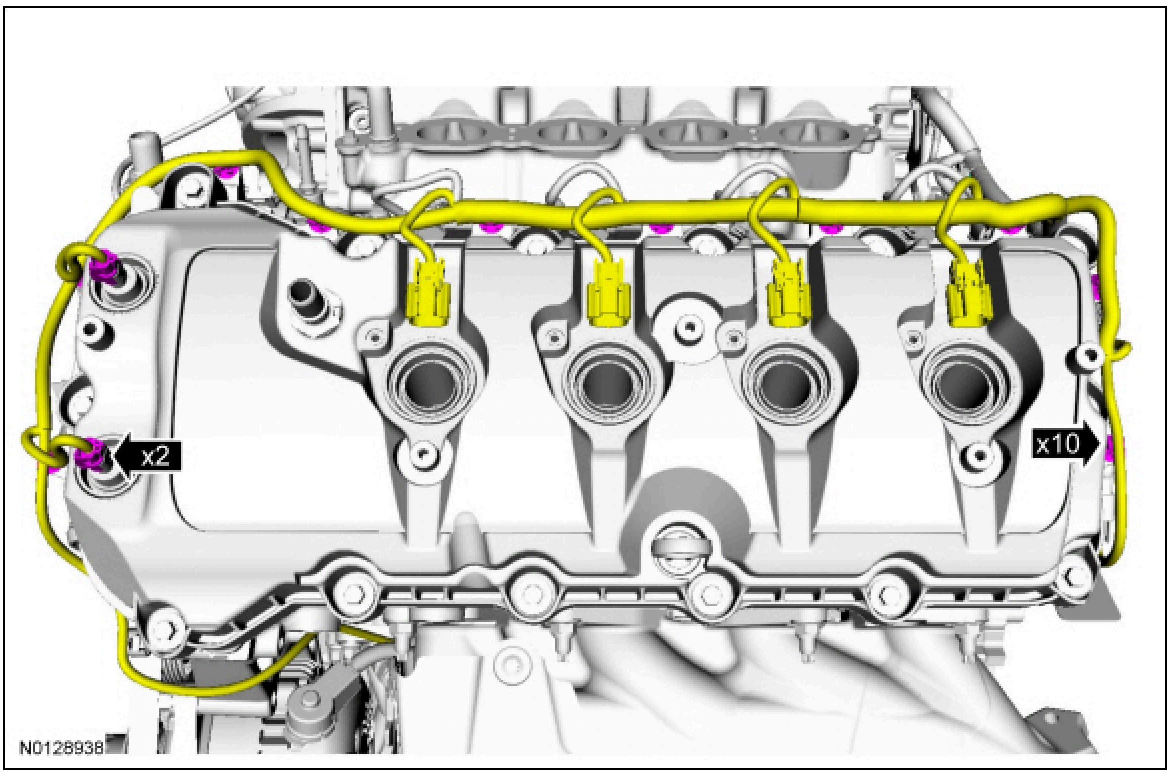
120. Install the RH intake and exhaust CMP sensors and the 2 bolts.
    - Tighten to 10 Nm (89 lb-in).
    - Connect the RH intake and exhaust CMP sensor electrical connectors.



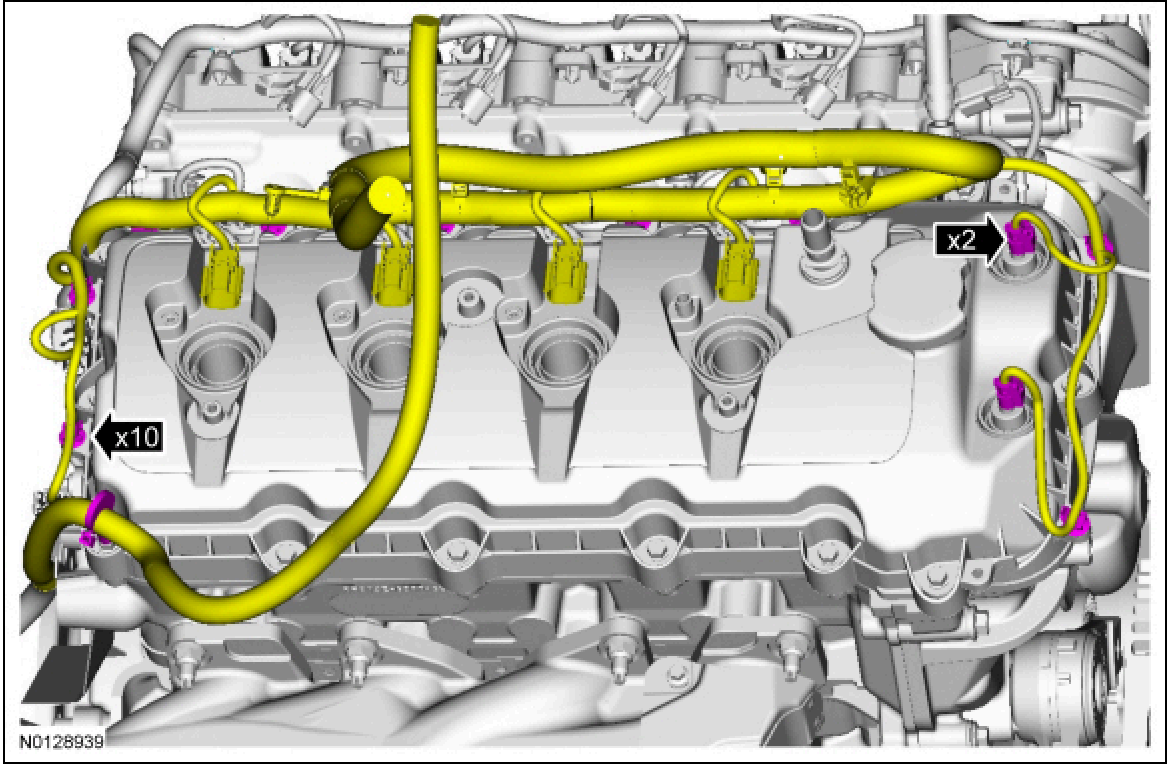
121. Install the heat shield, the bolt and the stud bolt on the RH cylinder head.
    - Tighten to 10 Nm (89 lb-in).



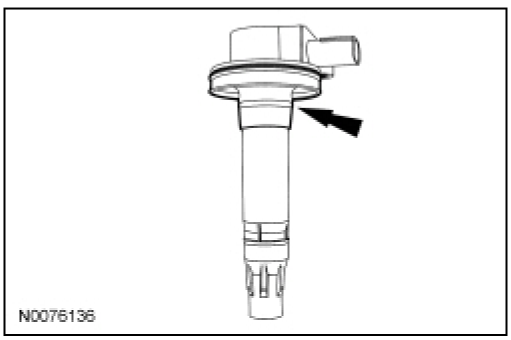
122. Connect the 2 VCT variable force solenoid electrical connectors.
    - Attach the 10 wiring harness retainers to the LH valve cover.



123. Connect the 2 Variable Camshaft Timing (VCT) variable force solenoid electrical connectors.
    - Attach the 10 wiring harness retainers to the RH valve cover.

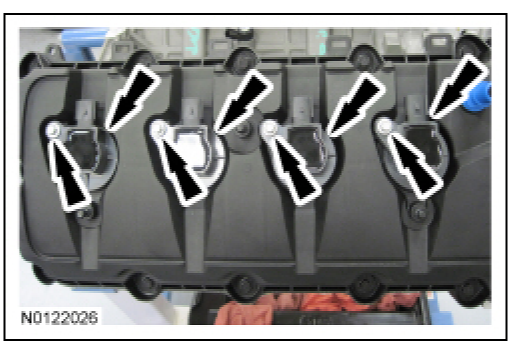


124. Inspect the coil seals for rips, nicks or tears. Remove and discard any damaged coil seals.
    - To install, slide the new coil seal onto the coil until it is fully seated at the top of the coil.



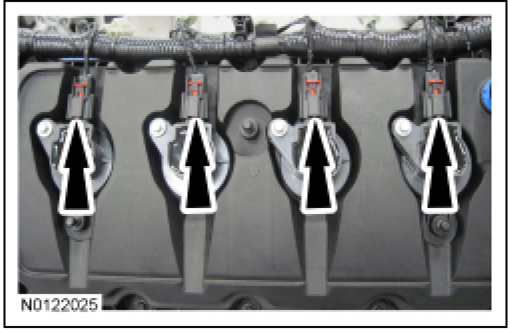
125. **NOTE:** *Apply a small amount of dielectric grease to the inside of the ignition coil-on-plug boots before attaching to the spark plugs. RH shown, LH similar.*

    Install the 8 ignition coils and 8 bolts.
    - Tighten to 6 Nm (53 lb-in).

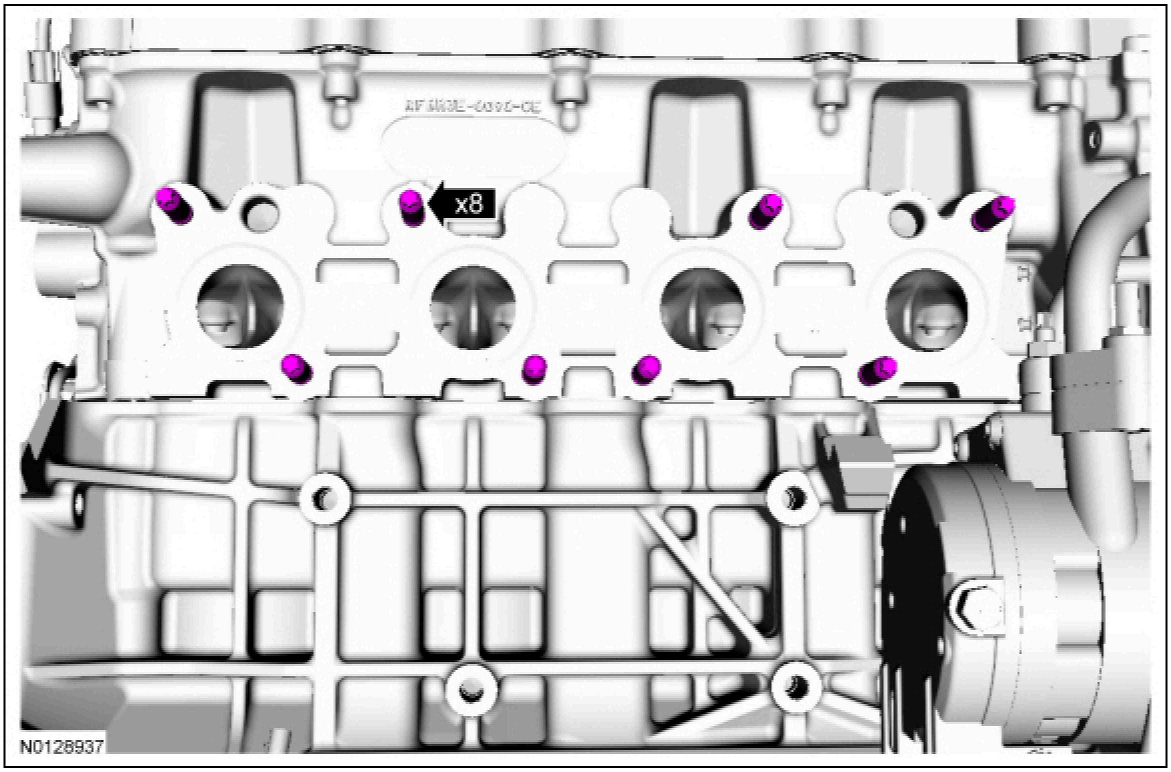


126. **NOTE:** *RH shown, LH similar.*

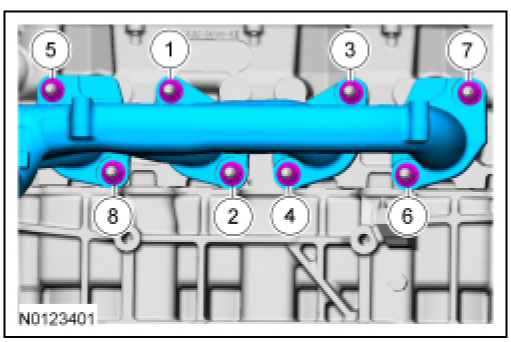
    Connect the 8 ignition coil electrical connectors.



127. Install 8 new RH exhaust manifold studs.
    - Tighten to 25 Nm (18 lb-ft).

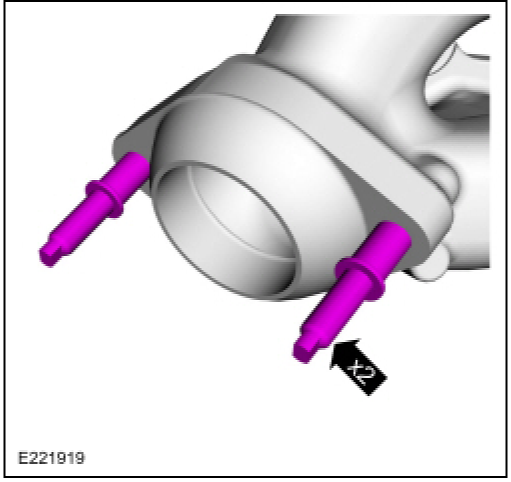


128. Install a new gasket, the RH exhaust manifold and 8 new nuts.
    - Tighten the nuts in the sequence shown in 2 stages.
        - Stage 1: Tighten to 24 Nm (18 lb-ft).
        - Stage 2: Tighten to 32 Nm (24 lb-ft).

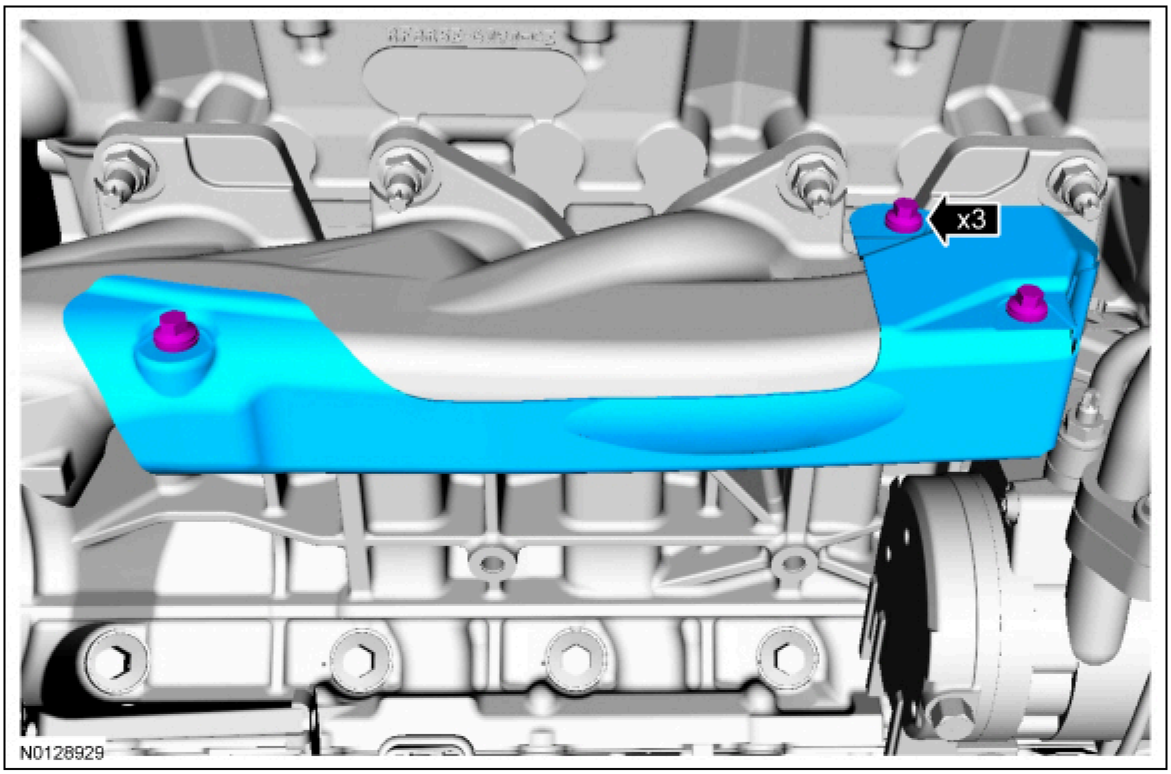


129. **NOTE:** *Some replacement exhaust manifolds do not come with studs*.

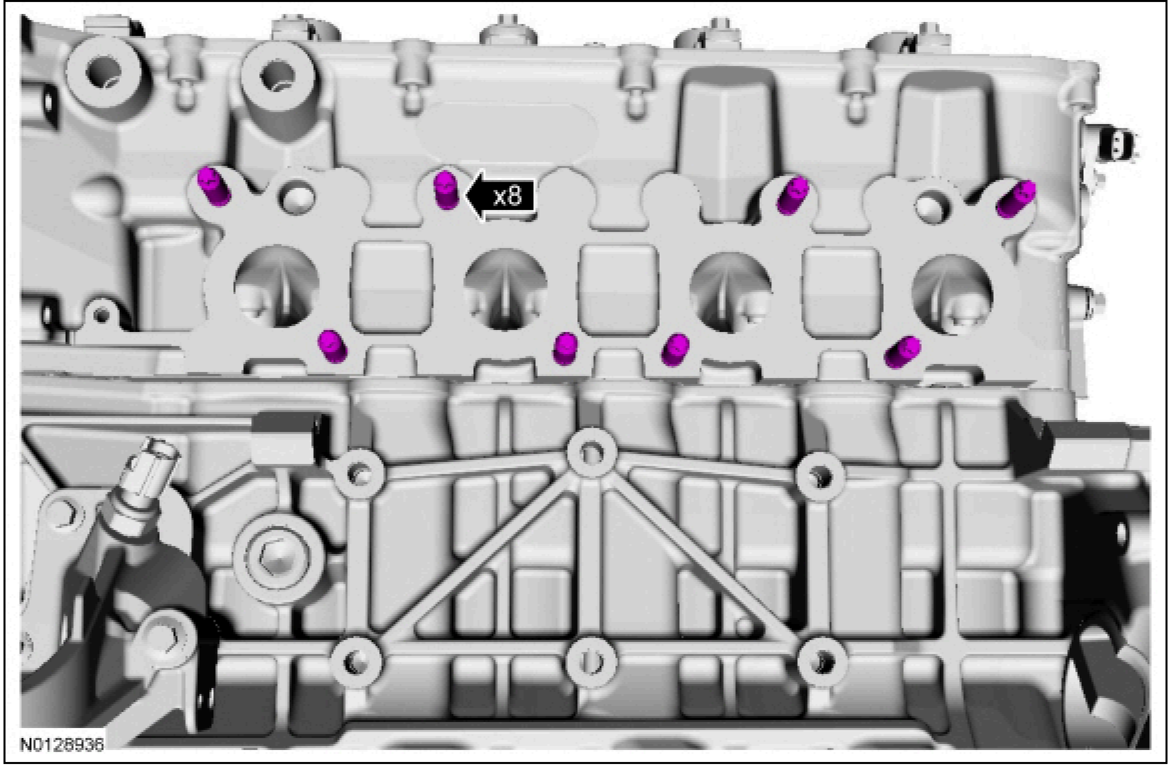
    If necessary, install new studs.
    - Tighten to 40 Nm (30 lb-ft)



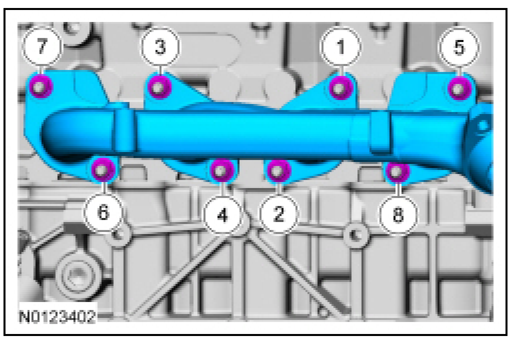
130. Install the RH exhaust manifold heat shield and the 3 bolts.
    - Tighten to 12 Nm (106 lb-in).



131. Install 8 new LH exhaust manifold studs.
    - Tighten to 25 Nm (18 lb-ft).

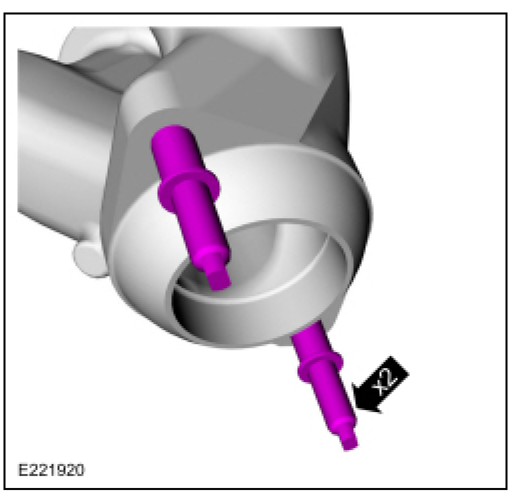


132. Install a new gasket, the LH exhaust manifold and 8 new nuts.
- Tighten in the sequence shown in 2 stages.
  - Stage 1: Tighten to 24 Nm (18 lb-ft).
  - Stage 2: Tighten to 32 Nm (24 lb-ft).

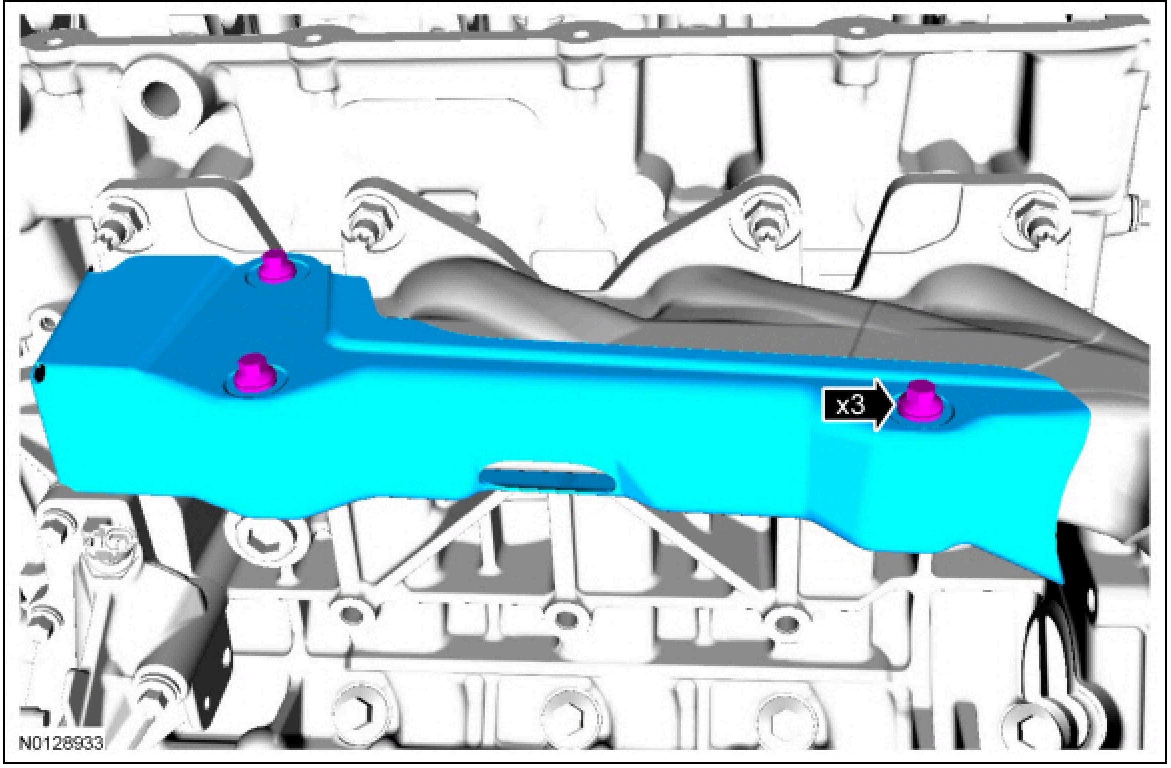


133. **NOTE:** *Some replacement exhaust manifolds do not come with studs*.

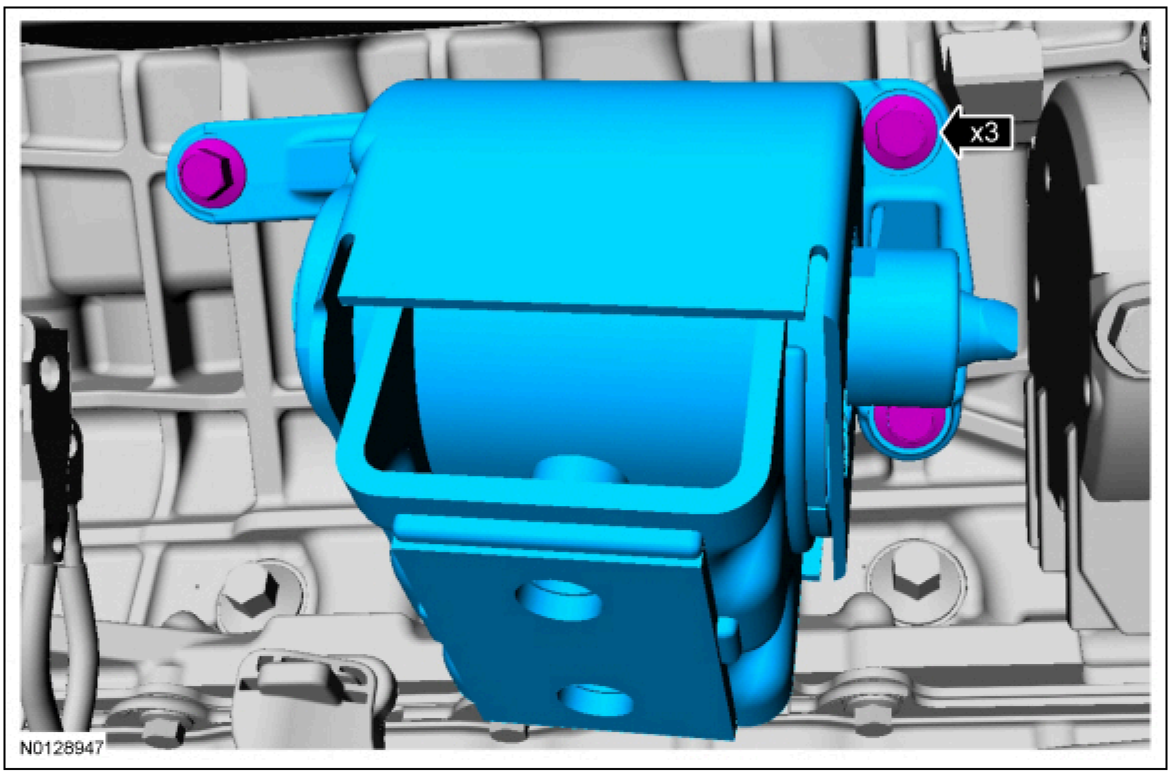
If necessary, install new studs.
- Tighten to 40 Nm (30 lb-ft)



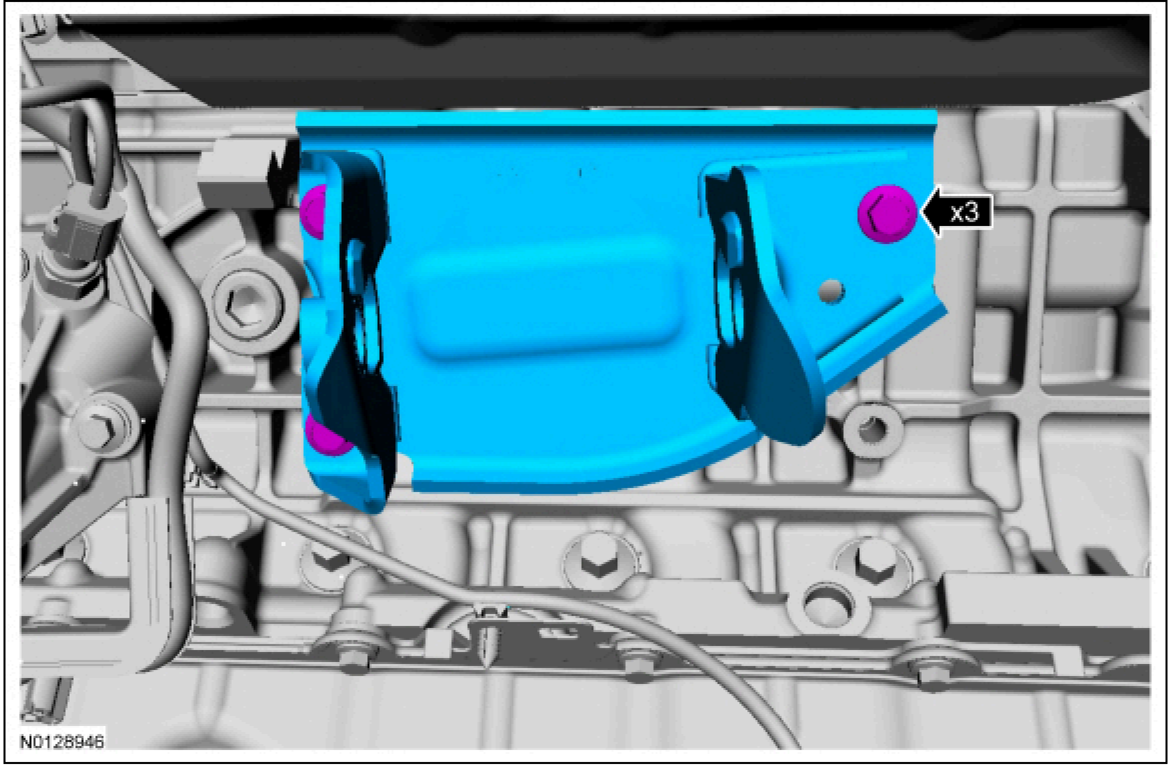
134. Install the LH exhaust manifold heat shield and the 3 bolts.
- Tighten to 12 Nm (106 lb-in).



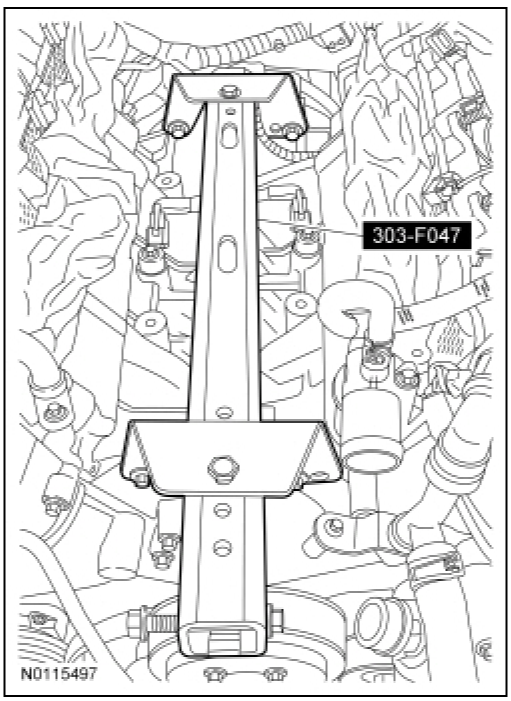
135. Install the RH engine support insulator and the 3 bolts.
    - Tighten to 63 Nm (46 lb-ft).



136. Install the LH engine support insulator bracket and the 3 bolts.
    - Tighten to 63 Nm (46 lb-ft).



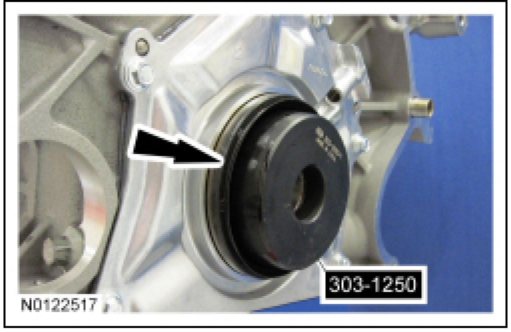
137. Install the Engine Lifting Bracket



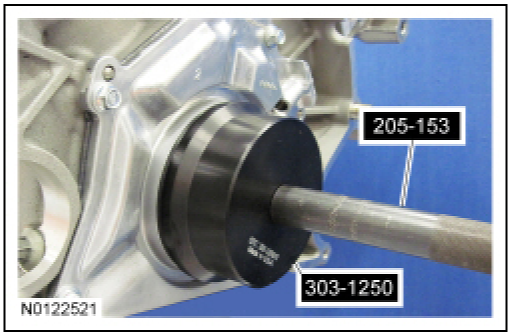
138. Using the Floor Crane, remove the engine from the work stand.

139. **NOTE:** *Lubricate the seal lips and bore with clean engine oil prior to installation.*

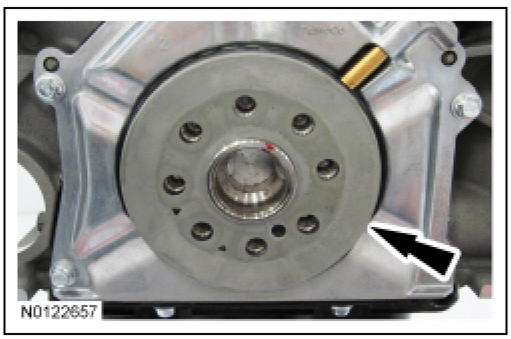
Position the Rear Main Seal Installer onto the end of the crankshaft and slide a new crankshaft rear seal onto the tool.



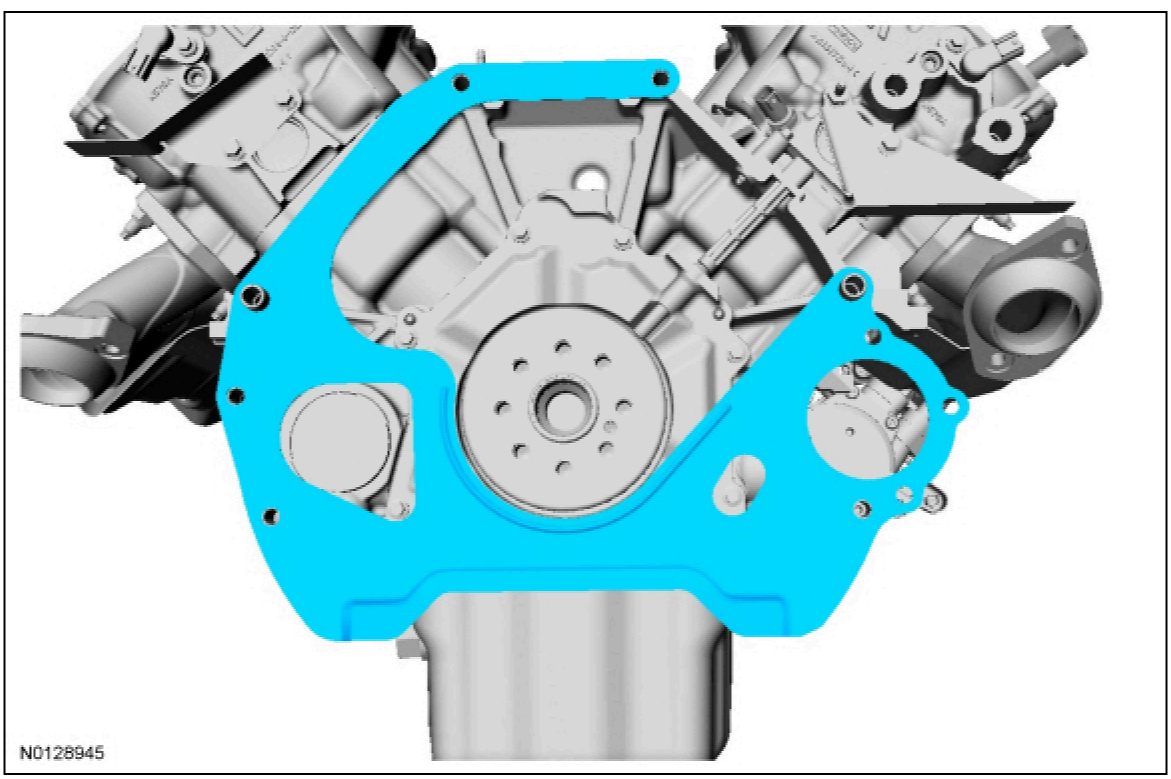
140. Using the Rear Main Seal Installer and Handle, install the new crankshaft rear seal.



141. Install the crankshaft sensor ring.



142. Install the engine separator plate.



143. **NOTE:** *The flexplate bolts are torque-to-yield design and are not reusable*.

Install the engine separator plate, the flexplate and 8 new bolts.

- Tighten in the sequence shown in 2 stages.
  - Stage 1: Tighten to 20 Nm (177 lb-in).
  - Stage 2: Tighten an additional 60 degrees.